

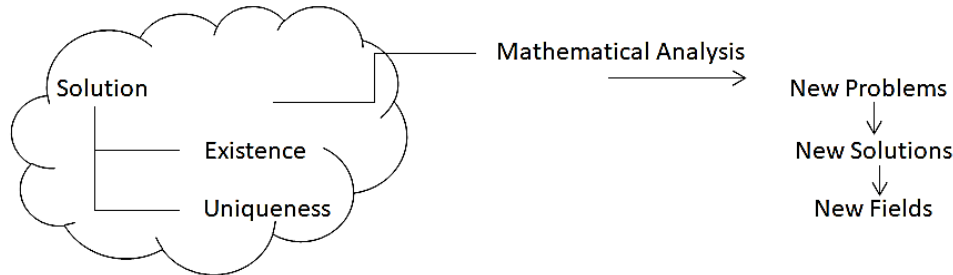
LECTURE NO. 01

Mathematical Modelling: It is the approximation of objects with geometrical objects.

Mathematical Formulation: In this step we drive the equation corresponds to the given phenomenon. It is based on existing law i.e. Newton's law of motion, laws of thermodynamics. If laws are not available for given existing problem, then we go for experiments.

Example: $F_e = G \frac{m_e m_s}{(r_{es})^2} \leftrightarrow \frac{d^2 r}{dt^2} = G \frac{m}{r}$

LECTURE NO. 02



Example: Heat flow problem $\rightarrow$ Formulation $\rightarrow$ PDEs $\rightarrow$ Solutions $\rightarrow$ Expansion of function terms of 'sine' and 'cosine' (Fourier analysis).

Physical Interpretation: E.g. $x = \frac{dx}{dt} \leftrightarrow x(t) = Ae^t \quad \therefore A \in R$

LECTURE NO. 03

Vibrating String Equation:

$$\frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2}$$

Here we'll try to derive the above given equation. So,

Let ' Δs ' represents the elements of arc of string and ' T ' is tension at ' x ' and ' $x + \Delta x$ ' (constant).

$$\text{Net upward force} = T \sin \theta_2 - T \sin \theta_1$$

For small angles; $\sin \theta \approx \tan \theta \quad \therefore \sin 0 = \tan 0 = 0$

So $\text{Net force} = T \tan \theta_2 - T \tan \theta_1 \quad \dots (1)$

By newton second law of motion

$$\text{Force} = ma = (\text{mass of } \Delta s)(\text{acceleration of } \Delta s)$$

$$\text{Force} = (\mu \Delta s) \left(\frac{\partial^2 y}{\partial t^2} + \varepsilon \right) \quad \dots (2) \quad \therefore \mu \text{ is linear mass density}$$

Note: Here acceleration is in derivative form and while discussing the acceleration for a segment (large number of particles), there will be some approximation/error. That's why we are using ' ϵ ' as approximation symbol.

By comparing equation 1 and 2, we have

$$T \tan \theta_2 - T \tan \theta_1 = (\mu \Delta s) \left(\frac{\partial^2 y}{\partial t^2} + \varepsilon \right)$$

$$T [\tan \theta_2 - \tan \theta_1] = (\mu \Delta s) \left(\frac{\partial^2 y}{\partial t^2} + \varepsilon \right)$$

As $\tan \theta = \frac{\Delta y}{\Delta x} = \frac{dy}{dx}$, so above equation become

$$T \left[\left(\frac{dy}{dx} \right)_{x+\Delta x} - \left(\frac{dy}{dx} \right)_x \right] = (\mu \Delta s) \left(\frac{\partial^2 y}{\partial t^2} + \varepsilon \right)$$

Note: Here ' $x + \Delta x$ and x ' are 'final and initial points' respectively.

For small vibrations; $\Delta s = \Delta x \quad \leftrightarrow \quad \mu \Delta s = \mu \Delta x$

By using this relation, we get

$$T \left[\left(\frac{dy}{dx} \right)_{x+\Delta x} - \left(\frac{dy}{dx} \right)_x \right] = \mu \Delta x \left(\frac{\partial^2 y}{\partial t^2} + \varepsilon \right)$$

Dividing by Δx on both sides

$$T \left[\frac{\left(\frac{dy}{dx} \right)_{x+\Delta x} - \left(\frac{dy}{dx} \right)_x}{\Delta x} \right] = \frac{\mu \Delta x}{\Delta x} \left(\frac{\partial^2 y}{\partial t^2} + \varepsilon \right)$$

$$\frac{T}{\mu} \left[\frac{\left(\frac{dy}{dx} \right)_{x+\Delta x} - \left(\frac{dy}{dx} \right)_x}{\Delta x} \right] = \left(\frac{\partial^2 y}{\partial t^2} + \varepsilon \right)$$

For $\Delta x \rightarrow 0$ and $\varepsilon \rightarrow 0$, it becomes

$$\frac{T}{\mu} \left[\frac{\partial}{\partial x} \left(\frac{\partial y}{\partial x} \right) \right] = \frac{\partial^2 y}{\partial t^2}$$

$$\frac{T}{\mu} \left(\frac{\partial^2 y}{\partial x^2} \right) = \frac{\partial^2 y}{\partial t^2}$$

Here $\frac{T}{\mu} = a^2$, so equation becomes

$$a^2 \left(\frac{\partial^2 y}{\partial x^2} \right) = \frac{\partial^2 y}{\partial t^2}$$

Or it can be written as

$$\frac{\partial^2 y}{\partial t^2} = a^2 \left(\frac{\partial^2 y}{\partial x^2} \right)$$

This relation is known as Vibrating string equation.

LECTURE NO. 04

Boundary Conditions for Vibrating String Equation:

Statement: Write boundary conditions for a vibrating string of length ' L ' for which i) Ends $x = 0$ and $x = L$ are fixed. ii) Initial shape of string is $f(x)$. iii) Initial velocity distribution is $g(x)$. iv) Displacement at any instant ' t ' is bounded.

Solution:

i) String is fixed at $x = 0$ and $x = L$, then its displacement at

$$\text{starting point: } y(0, t) = 0 = \text{end point: } y(L, t) \quad \forall \quad t > 0$$

Note: above relation is in " $\text{function (displacement, time)} = y(x, t)$ " form. At starting point " $x = 0$ " and at final point " $x = L$ ".

ii) Initial shape at $t = 0$ is given as

$$y(x, 0) = f(x)$$

Note: Just put $t = 0$ in " $y(x, t)$ " relation and here $f(x)$ is just an expression.

iii) Initial velocity at time ' t ' is given as

$$\frac{\partial y}{\partial t} = y_t(x, t)$$

At $t = 0$

$$y_t(x, 0) = g(x) \quad \forall \quad 0 < x < L$$

Note: Here velocity is in derivative form and $g(x)$ is just an expression.

iv) Displacement is bounded: It implies that there exist some finite real number ($M < \infty$) such that

$$|y(x, t)| < M \quad \forall \quad 0 < x < L \quad \text{and} \quad t > 0$$

These are some examples of boundary conditions applied on vibrating string equation.

DEDICATED TO UNKNOWN STUDENTS WHO ARE OUR FUTURE HEROES.

LECTURE NO. 05

Heat Flux across the Plane:

Suppose planes 'I' and 'II' are parallel and at distance 'Δn' apart. Let
u = temp of plane I And *u* + Δ*u* = temp of plane II

Heat flows from higher temp plane to lower temp plane.

Heat flux is defined as; Heat per unit area per unit time. Mathematically

$$\text{Heat flux} \propto \frac{\text{difference of temperature}}{\text{distance between planes}}$$
$$\text{Heat flux form I to II} = k \frac{(u) - (u + \Delta u)}{\Delta n} \Rightarrow -k \frac{\Delta u}{\Delta n}$$

Where 'k' is constant of proportionality and is called thermal conductivity.

Thermal conductivity is defined as; Measure of how well a material can conduct or transfer heat.

Note: As in this case heat flows from plane 'I to II' (lower temp to higher temp), that's why we have the negative value of heat flux. ∴ Δ*u* > 0 (Heat value will be positive) if heat flows from plane 'II to I' (higher temp to low temp) and vice versa.

For limiting case: If Δ*n* → 0 and Δ*u* → 0 then

$$\text{Heat flow across plane I} = -k \frac{\partial u}{\partial n}$$

In vector form

$$\text{Heat flow across plane I} = \left\{ -k \frac{\nabla u}{\nabla n} \right\}$$

Note: As Δ*n* → 0 (distance between planes approaches to zero), so we can write it as plane 'I' only.

LECTURE NO. 06

Partial Differential Equations (Definition and Related Terms):

Definition: It is an equation containing unknown functions of two or more variables and partial derivatives w.r.to these variables. Example

$$\frac{\partial^2 u}{\partial x \partial y} = 2x - y \quad \text{order} = 2$$

Order: It is the order of highest derivative involved in PDE.

Solution: It is a function which satisfy the given DE. Example:

$$u = x^2y - \frac{1}{2}xy^2$$
$$\frac{\partial u}{\partial y} = x^2 - xy \quad \text{1st derivative w.r.to 'y'}$$
$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right) = 2x - y \quad \text{2nd derivative w.r.to 'x'}$$

Similarly; *u* = *x*²*y* − ¹/₂*xy*² + *F*(*x*) + *G*(*y*) is also a solution. Here '*F*(*x*) and *G*(*x*)' are arbitrary functions.

Particular Solution: It is obtained from the general solution by particular choice of arbitrary functions.
Example:

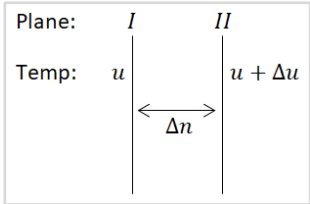
$$u = x^2y - \frac{1}{2}xy^2 + 2\sin x + 3y^4 - 5$$

Singular Solution: It cannot be obtained from the general solution by choosing arbitrary functions.

LECTURE NO. 07

Example: *xy*' = √*y* => *y* = ¹/₄(*lncx*)² (General Solution—Here 'c' is arbitrary constant).

But '*y* = 0' is also a solution of given DE. As it is not obtained from general solution, that's why it is called singular solution. (∄ *c* ∈ *R* such that *y* = 0)



LECTURE NO. 08

Linear PDEs and Their Classification:

General form of linear partial differential equation of order 2 in two independent variables is of the form

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} + D \frac{\partial u}{\partial x} + E \frac{\partial u}{\partial y} + Fu = G$$

Here A, B, C, D, E, F and G may depend on ' x ' or ' y ' only but not on both ' x ' and ' y '.
If at least one of these (A, B, C, D, E, F and G) is function of ' x ' and ' y ', then it is non-linear.
If $G = 0$, then it is homogeneous.

Example:

$$x^2 \frac{\partial^3 u}{\partial y^3} = y^3 \frac{\partial^2 u}{\partial x^2} \qquad \qquad \qquad (order = 3, linear)$$

Example:

$$\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = 1 \qquad \qquad \qquad (order = 1, non - linear)$$

LECTURE NO. 09

Topic Continue...

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} + D \frac{\partial u}{\partial x} + E \frac{\partial u}{\partial y} + Fu = G$$

It is defined as

- i) Elliptical if $B^2 - 4AC < 0$
- ii) Hyperbolic if $B^2 - 4AC > 0$
- iii) Parabolic if $B^2 - 4AC = 0$

Example:

$$x \frac{\partial^2 u}{\partial x^2} + y \frac{\partial^2 u}{\partial y^2} + 3y^2 \frac{\partial u}{\partial x} = 0$$

Here $A = x, b = 0$ and $c = y$. Now

$$B^2 - 4AC = -4xy \begin{cases} xy > 0, \text{ then it is elliptic} \\ xy < 0, \text{ then it is hyperbolic} \\ xy = 0, \text{ then it is parabolic} \end{cases}$$

LECTURE NO. 10

Show that $u(x, t) = e^{-8t} \sin 2x$, is a solution to BVP (Boundary Value Problem):

I) $\frac{\partial u}{\partial t} = 2 \frac{\partial^2 u}{\partial x^2}$, II) $u(0, t) = u(\pi, t) = 0$, III) $u(x, 0) = \sin 2x$

Solution:

- II) $u(x, t) = e^{-8t} \sin 2x \Rightarrow u(0, t) = e^{-8t} \sin(0) = 0$
- II) $u(x, t) = e^{-8t} \sin 2x \Rightarrow u(\pi, t) = e^{-8t} \sin(2\pi) = 0$
- III) $u(x, t) = e^{-8t} \sin 2x \Rightarrow u(x, 0) = e^{-8(0)} \sin 2x = \sin 2x$

Now, try to calculate the values for condition 'I'

$$\frac{\partial u}{\partial t} = 2 \frac{\partial^2 u}{\partial x^2} \qquad \dots \dots (1)$$

Taking derivative of equation 1 w.r.to ' t '

$$\frac{\partial u}{\partial t} = -8e^{-8t} \sin 2x$$

Taking derivative of equation 1 w.r.to 'x'

$$\frac{\partial u}{\partial x} = 2e^{-8t} \cos 2x$$

Taking 2nd derivative, we have

$$\frac{\partial^2 u}{\partial x^2} = -4e^{-8t} \sin 2x$$

Now, by putting the calculated values in equation 1 and checking, we get

$$\begin{aligned} \frac{\partial u}{\partial t} &= 2 \frac{\partial^2 u}{\partial x^2} \\ \frac{\partial u}{\partial t} &= -8e^{-8t} \sin 2x \Rightarrow 2(-4e^{-8t} \sin 2x) = 2 \frac{\partial^2 u}{\partial x^2} \end{aligned}$$

Or simply

$$\frac{\partial u}{\partial t} = 2 \frac{\partial^2 u}{\partial x^2}$$

Hence proved('I') that the given function " $u(x, t) = e^{-8t} \sin 2x$ " is solution to given BVP.

LECTURE NO. 11

Show that $v = F(y - 3x)$, where 'F' is arbitrary differentiable function.

I) The general solution of equation is $\frac{\partial v}{\partial x} + 3 \frac{\partial v}{\partial y} = 0$.

II) Also find its particular solution if $v(0, y) = 4 \sin y$

Solution:

I) Given that $v = F(y - 3x)$ (1)

Let / say $v(x, y) = F(u)$ Where $u = y - 3x$
 $v = F(u)$ and $u = u(x, y)$

By applying chain rule w.r.to 'x'

$$\frac{\partial v}{\partial x} = \frac{\partial v}{\partial u} \cdot \frac{\partial u}{\partial x} = F'(u)(-3) \Rightarrow -3F'(u)$$

By applying chain rule w.r.to 'y'

$$\frac{\partial v}{\partial y} = \frac{\partial v}{\partial u} \cdot \frac{\partial u}{\partial y} = F'(u)(1) \Rightarrow F'(u)$$

Now by putting value in equation

$$\frac{\partial v}{\partial x} + 3 \frac{\partial v}{\partial y} = -3F'(u) + 3F'(u)$$

By solving, we have

$$\frac{\partial v}{\partial x} + 3 \frac{\partial v}{\partial y} = 0$$

Hence, the general solution of equation proved.

II) Now, also given that

$$v(x, y) = F(y - 3x)$$

Given condition

$$v(0, y) = F(y) = 4 \sin y$$

It implies

$$v(x, y) = F(y - 3x) = 4 \sin(y - 3x)$$

This is our required particular solution.

LECTURE NO. 12

Solving PDEs by the Method of ODEs:

Question:

$$t \frac{\partial^2 u}{\partial x \partial t} + 2 \frac{\partial u}{\partial x} = 2tx$$

Solution: Given that

$$\begin{aligned} t \frac{\partial^2 u}{\partial x \partial t} + 2 \frac{\partial u}{\partial x} &= 2tx \\ t \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial t} \right) + 2 \frac{\partial}{\partial x} u &= 2tx \\ \frac{\partial}{\partial x} \left\{ t \frac{\partial u}{\partial t} + 2u \right\} &= 2tx \end{aligned}$$

Integrating w.r.to 'x'

$$\begin{aligned} \int \frac{\partial}{\partial x} \left\{ t \frac{\partial u}{\partial t} + 2u \right\} dx &= \int 2tx \, dx \\ t \frac{\partial u}{\partial t} + 2u &= x^2 t + G(t) \end{aligned}$$

Dividing by 't' on both sides

$$\begin{aligned} \frac{\partial u}{\partial t} + \frac{2}{t}u &= x^2 + \frac{G(t)}{t} \\ \text{Integrating Factor} = I.F &= e^{\int \frac{2}{t} dt} \\ I.F &= e^{2 \ln t} = e^{\ln t^2} = t^2 \end{aligned}$$

Required solution is

$$I.F \times u = \int (R.H.S \times I.F) dt + F(x)$$

By putting values

$$\begin{aligned} t^2 u &= \int \left(\left(x^2 + \frac{G(t)}{t} \right) \times t^2 \right) dt + F(x) \\ t^2 u &= \int \left(x^2 t^2 + \frac{G(t)}{t} t^2 \right) dt + F(x) \\ t^2 u &= x^2 \frac{t^3}{3} + \int t.G(t) dt + F(x) \\ t^2 u &= x^2 \frac{t^3}{3} + H(t) + F(x) \end{aligned}$$

LECTURE NO. 13

General Solution for Solving PDEs:

Question

$$\frac{\partial^2 u}{\partial x^2} + 3 \frac{\partial^2 u}{\partial x \partial t} + 2 \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{..... (1)}$$

Solution: Let

$$u = e^{ax+by}$$

Taking 1st and 2nd derivatives w.r.to 'x'

$$\begin{aligned} \frac{\partial u}{\partial x} &= e^{ax+by} \frac{\partial}{\partial x} (ax + by) = au \\ \frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) = \frac{\partial}{\partial x} (au) = a \frac{\partial u}{\partial x} = a^2 u \end{aligned}$$

Similarly taking derivatives w.r.to 'y'

$$\frac{\partial^2 u}{\partial y^2} = b^2 u$$

And

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right) = \frac{\partial}{\partial x} (bu) = b \frac{\partial u}{\partial x} = abu$$

Putting values in equation 1, we have

$$\begin{aligned}a^2u + 3abu + 2b^2u &= 0 \\(a^2 + 3ab + 2b^2)u &= 0 \\(a + 2b)(a + b) &= 0\end{aligned}$$

By solving, we get

$$a = -2b \quad ; \quad a = -b$$

For $a = -2b$

$$u_1 = e^{-2bx+by} = e^{b(y-2x)}$$

For $a = -b$

$$u_2 = e^{-bx+by} = e^{b(y-x)}$$

Given PDE is homogeneous, therefore by superposition principle

$$u = \alpha u_1 + \beta u_2$$

By putting values

$$u = \alpha e^{b(y-2x)} + \beta e^{b(y-x)}$$

Let

$$e^{b(y-2x)} = G(y - 2x) \quad \text{and} \quad e^{b(y-x)} = H(y - x)$$

So, above equation becomes

$$u = G(y - 2x) + H(y - x)$$

Required general solution of given PDE.

LECTURE NO. 14

Solving PDEs by Separation of Variables:

Question:

$$\frac{\partial u}{\partial x} = 4 \frac{\partial u}{\partial y} \quad \text{subject to boundry conditio } u(0, y) = 8e^{-3y}$$

Solution: Given equation

$$\frac{\partial u}{\partial x} = 4 \frac{\partial u}{\partial y} \quad \dots (1)$$

Let

$$u(x, y) = X(x) Y(y) = XY \qquad \text{Be its solution... (i)}$$

Taking derivative w.r.to x & y

$$\frac{\partial u}{\partial x} = X'Y \quad \text{and} \quad \frac{\partial u}{\partial y} = XY'$$

Putting values in equation 1, we have

$$\begin{aligned}X'Y &= 4XY' \\ \frac{X'}{4X} &= \frac{Y'}{Y} \qquad \dots (2)\end{aligned}$$

As

$$X(x) = X, Y(y) = Y$$

Here X and Y are independent variables. Each side of equation 2 must be a constant say ' c '. So,

$\frac{X'}{4X} = c$	$\frac{Y'}{Y} = c$
$X' = 4cX$	$Y' = cY$
$\frac{dX}{dx} = 4cX$	$\frac{dY}{dy} = cy$
$\frac{dX}{X} = 4cdx$	$\frac{dY}{Y} = cdy$

By integrating

$\int \frac{dX}{X} = 4c \int dx$	$\int \frac{dY}{Y} = c \int dy$
$\ln X = 4cx + A$	$\ln Y = cy + B$
$X = e^{4cx+A}$	$Y = e^{cy+B}$
$X = e^A e^{4cx}$	$Y = e^B e^{cy}$
$X = k_1 e^{4cx}$	$Y = k_2 e^{cy}$

Now by putting values in equation (i)

$$u(x,y)=XY=k_1k_2e^{4cx+cy} \dots\dots (3)$$

By using boundary condition $u(0,y)=8e^{-3y}$ in equation 3, we have

$$u(0,y)=k e^{cy}=8e^{-3y}$$

Here $k=8$ and $c=-3$,

Hence

$$u(x,y)=8e^{-3(4x+y)}=8e^{-12x-3y}$$

LECTURE NO. 15

Heat Conduction Equation and its Physical Interpretation:

Question:

$$\frac{\partial u}{\partial t}=2\frac{\partial^2 u}{\partial x^2} \dots (1) \qquad \therefore 0 < x < 3, t > 0$$

Given that (boundary values)

$$u(0,t)=u(3,t)=0 \qquad \dots (i),$$

$$u(x,0)=5\sin 4\pi x-3\sin 8\pi x+2\sin 10\pi x \qquad \dots (ii) \therefore |u(x,t)| < M < \infty$$

Solution: Let

$$u(x,t)=X(x).T(t)=XT \dots (z)$$

Taking derivative w.r.to 't'

$$\frac{\partial u}{\partial t}=XT'$$

Taking 1st derivative w.r.to 'x'

$$\frac{\partial u}{\partial x}=X'T$$

Taking 2nd derivative w.r.to 'x'

$$\frac{\partial^2 u}{\partial x^2}=X''T$$

By putting values equation 1 becomes

$$\begin{aligned} XT' &= 2X''T \\ \frac{X''}{X} &= \frac{T'}{2T} = -\lambda^2 \text{ (say)} \end{aligned}$$

So,

$$\begin{aligned} \frac{X''}{X} &= -\lambda^2 \\ X'' + \lambda^2 X &= 0 \dots (a) \end{aligned}$$

$$\text{Let } X = e^{mx}$$

$$X'' = m^2 e^{mx} = m^2 X$$

$$(a) \Rightarrow m^2 X + \lambda^2 X = 0$$

$$m = \pm \lambda i \text{ \& } X \neq 0$$

$$\therefore X(x) = \alpha_1 e^{\lambda i x} + \alpha_2 e^{-\lambda i x}$$

$$X(x) = \alpha_1 [\cos \lambda x + i \sin \lambda x] + \alpha_2 [\cos \lambda x - i \sin \lambda x]$$

$$X(x) = (\alpha_1 + \alpha_2) \cos \lambda x + i(\alpha_1 - \alpha_2) \sin \lambda x$$

$$X(x) = A_1 \cos \lambda x + B_1 \sin \lambda x$$

$$\frac{T'}{2T} = -\lambda^2$$

Taking derivative

$$\frac{dT}{dt} = -2\lambda^2 T$$

$$\frac{dT}{T} = -2\lambda^2 dt$$

$$\int \frac{dT}{T} = -2\lambda^2 \int dt$$

$$\ln T = -2\lambda^2 t + c$$

$$T = e^{-2\lambda^2 t + c}$$

$$T = e^c \cdot e^{-2\lambda^2 t}$$

$$T = c_1 e^{-2\lambda^2 t}$$

$$\text{Here } e^c = c_1$$

Here we were solving both functions side by side. Now

Putting values in equation (z), we get

$$u(x,t)=XT=e^{-2\lambda^2 t}(A_1\cos \lambda x+B_1\sin \lambda x) \dots (2)$$

Calculating (i) (boundary value) by putting values in equation 2,

$$u(0,t) = 0 = e^{-2\lambda^2t}(A\cos\lambda(0) + B\sin\lambda(0)) = e^{-2\lambda^2t}.A \Rightarrow A = 0 \quad \therefore e^{-2\lambda^2t} \neq 0$$

Equation 2 becomes

$$u(x,t) = XT = B.e^{-2\lambda^2t}\sin\lambda x \quad \therefore A = 0$$

Calculating second part of (i) by putting values in above relation

$$u(3,t) = 0 = e^{-2\lambda^2t}.B\sin3\lambda$$

Two Cases

1; if $B = 0 \Rightarrow u(x,t) = 0$ as $A = 0$ already

2; if $B \neq 0$ but $\sin3\lambda = 0 \Rightarrow 3\lambda = m\pi \Rightarrow \lambda = \frac{m\pi}{3} \quad \therefore m \in Z$

Hence

$$u(x,t) = Be^{-2\frac{m^2\pi^2}{9}t} \sin\left(\frac{m\pi}{3}x\right) \quad \dots (3)$$

Calculating (ii)(boundary value)

$$u(x,0) = 5\sin4\pi x - 3\sin8\pi x + 2\sin10\pi x \quad \dots (b)$$

By applying superposition principle on equation 3,

$$u(x,t) = B_1e^{-2\frac{m_1^2\pi^2}{9}t} \sin\left(\frac{m_1\pi}{3}x\right) + B_2e^{-2\frac{m_2^2\pi^2}{9}t} \sin\left(\frac{m_2\pi}{3}x\right) + B_3e^{-2\frac{m_3^2\pi^2}{9}t} \sin\left(\frac{m_3\pi}{3}x\right) \quad \dots (4)$$

For

$$u(x,0) = B_1 \sin\left(\frac{m_1\pi}{3}x\right) + B_2 \sin\left(\frac{m_2\pi}{3}x\right) + B_3 \sin\left(\frac{m_3\pi}{3}x\right) \quad \dots (c)$$

By comparing b & C, we have

$$B_1 = 5, B_2 = -3, B_3 = 2, m_1 = 12, m_2 = 24 \text{ \& } m_3 = 30$$

Finally by putting calculated values in equation 4, we get

$$u(x,t) = 5e^{-32\pi^2t} \sin4\pi x - 3e^{-128\pi^2t} \sin8\pi x + 2e^{-200\pi^2t} \sin10\pi x$$

LECTURE NO. 16

Motivation Behind Fourier Series:

$$\frac{\partial u}{\partial t} = 2 \frac{\partial^2 u}{\partial x^2} \quad \dots (1) \quad \therefore 0 < x < 3, t > 0$$

Given that (boundary values)

$$u(0,t) = u(3,t) = 0 \quad \dots (i),$$

$$u(x,0) = 5\sin4\pi x - 3\sin8\pi x + 2\sin10\pi x \quad \dots (ii) \therefore |u(x,t)| < M < \infty$$

Solution: If

$$u(x,0) = f(x) = ? \quad \dots (1)$$

As we know from previous lecture

$$u(x,t) = B_1e^{-2\frac{m_1^2\pi^2}{9}t} \sin\left(\frac{m_1\pi}{3}x\right) + B_2e^{-2\frac{m_2^2\pi^2}{9}t} \sin\left(\frac{m_2\pi}{3}x\right) + \dots = \sum_{m=1}^{\infty} B_m e^{-2\frac{m^2\pi^2}{9}t} \sin\left(\frac{m\pi x}{3}\right)$$

Calculating (1)

$$f(x) = u(x,0) = \sum_{m=1}^{\infty} B_m e^{-2\frac{m^2\pi^2(0)}{9}} \sin\left(\frac{m\pi x}{3}\right) = \sum_{m=1}^{\infty} B_m e^0 \sin\left(\frac{m\pi x}{3}\right)$$

As $e^0 = 1$

$$f(x) = u(x,0) = \sum_{m=1}^{\infty} B_m \sin\left(\frac{m\pi x}{3}\right)$$

LECTURE NO. 17

Periodic & Piecewise Continuous Functions:

Definition: A function $f(x)$ is said to have a period ' p ' if $\forall x$, we have $f(x+p) = f(x)$, and ' $p > 0$ '. Further if $p > 0$ is least then it is said to be period of $f(x)$.

Example: $\sin x = \sin(x + 2\pi) = \sin(x + 4\pi) = \sin(x + 6\pi) = \dots$

But ' 2π ' is least $\Rightarrow p = 2\pi$ is a period of ' $\sin x$ '.

In general; for

$$\begin{aligned}\sin nx &= \sin(nx + 2\pi) \\ \sin n(x + 0) &= \sin n\left(x + \frac{2\pi}{n}\right) \\ \text{Period} = p &= \frac{2\pi}{n}\end{aligned}$$

Piecewise Continuous Functions:

$$f: [a, b] \rightarrow R$$

It is a function that has at most a finite number of finite discontinuities.

LECTURE NO. 18

Fourier Series:

Definition: let $f(x)$ be a periodic function of period ' $2l$ ' and defined on $(-l, l)$, then its Fourier series is defined as

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right) \right]$$

Where

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx, \quad b_n = \frac{1}{l} \int_{-l}^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx \quad \therefore n = 0, 1, 2, 3 \dots$$

These are called Fourier coefficient.

By changing the period, we can also define coefficients as

$$\begin{aligned}a_n &= \frac{1}{l} \int_c^{c+2l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx, \quad b_n = \frac{1}{l} \int_c^{c+2l} f(x) \sin\left(\frac{n\pi x}{l}\right) dx \quad \therefore n = 0, 1, 2, 3 \dots \\ \frac{a_0}{2} &= \frac{1}{2l} \int_{-l}^l f(x) dx \rightarrow \text{Average value of } f(x) \text{ in interval } (-l, l)\end{aligned}$$

Example: Function

$$\begin{aligned}f(x) &= x^2 \quad \& \quad \text{interval } (0, 2) \\ \int_0^2 x^2 dx &= \frac{1}{2} \left| \frac{x^3}{3} \right|_0^2 = \frac{1}{2} \cdot \frac{8}{3} = \frac{4}{3} \rightarrow \text{Average value in the interval } (0, 2)\end{aligned}$$

LECTURE NO. 19

Dirichlet Condition:

Fourier series corresponding to ' $f(x)$ ' is

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right) \right]$$

Question is; Series converge or diverge? If converge then converges to $f(x)$ or not.

Suppose that:

- i) $f(x)$ is defined and single valued on $(-l, l)$
 ii) $f(x)$ is periodic on $(-l, l)$ with period ' $2l$ '
 iii) $f(x)$ & $f'(x)$ are piecewise continuous, then the series

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right) \right] = \begin{cases} f(x) & \text{if 'x' is point of continuity} \\ \frac{f(x+0) + f(x-0)}{2} & \text{if 'x' is point of discontinuity} \end{cases}$$

LECTURE NO. 20

Evaluation of Fourier Series:

Theorem: If the series;

$$A + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right) \right]$$

Converges uniformly to $f(x)$ in $(-l, l)$, then for $n = 0, 1, 2, 3 \dots$ show that

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx \quad \dots (i)$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx \quad \dots (ii)$$

$$A = \frac{a_0}{2} \quad \dots (iii)$$

Solution: Given that

$$f(x) = A + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right) \right] \quad \dots (1)$$

Calculating i):

Multiplying (equation 1) both sides by $\cos\left(\frac{m\pi x}{l}\right)$ and integrating from $-l$ to l , we have

$$\int_{-l}^l f(x) \cos\left(\frac{m\pi x}{l}\right) dx = \int_{-l}^l A \cos\left(\frac{m\pi x}{l}\right) dx + \sum_{n=1}^{\infty} \left\{ a_n \int_{-l}^l \cos\left(\frac{n\pi x}{l}\right) \cos\left(\frac{m\pi x}{l}\right) dx + b_n \int_{-l}^l \sin\left(\frac{n\pi x}{l}\right) \cos\left(\frac{m\pi x}{l}\right) dx \right\} \dots (2)$$

$\downarrow$
(a)

$\downarrow$
(b)

$\downarrow$
(c)

Now we'll calculate the values for above functions.

$$(a); \quad \int_{-l}^l A \cos\left(\frac{m\pi x}{l}\right) dx = A \frac{l}{m\pi} \int_{-l}^l \left(\frac{m\pi}{l}\right) \cos\left(\frac{m\pi x}{l}\right) dx = \frac{Al}{m\pi} \left[\sin\left(\frac{m\pi x}{l}\right) \right]_{-l}^l = 0$$

$$(b); \quad \int_{-l}^l \cos\left(\frac{n\pi x}{l}\right) \cos\left(\frac{m\pi x}{l}\right) dx = \frac{1}{2} \int_{-l}^l \left\{ \cos\frac{(m+n)\pi x}{l} + \cos\frac{(m-n)\pi x}{l} \right\} dx = 0 \quad ; \quad \text{for } m \neq n$$

$\therefore$ Here we use the trigonometric relation; $\cos A \cos B = \frac{1}{2} [\cos(A+B) + \cos(A-B)]$

For $m = n$, we have

$$\int_{-l}^l \cos\left(\frac{n\pi x}{l}\right) \cos\left(\frac{m\pi x}{l}\right) dx = \frac{1}{2} \int_{-l}^l \cos^2\left(\frac{m\pi x}{l}\right) dx = \frac{1}{2} \int_{-l}^l \left\{ 1 + \cos 2\left(\frac{m\pi x}{l}\right) \right\} dx = \frac{1}{2} [x]_{-l}^l = l$$

$$(c); \quad \int_{-l}^l \sin\left(\frac{n\pi x}{l}\right) \cos\left(\frac{m\pi x}{l}\right) dx = \frac{1}{2} \int_{-l}^l \left\{ \sin\frac{(m+n)\pi x}{l} + \sin\frac{(m-n)\pi x}{l} \right\} dx = 0$$

$\therefore$ Here we use the trigonometric relation; $\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$

LECTURE NO. 21

By putting values in equation 2, we get

$$\int_{-l}^l f(x) \cos\left(\frac{m\pi x}{l}\right) dx = 0 + a_m l + 0 \Rightarrow a_n = \frac{1}{l} \int_{-l}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

This is our required solution for (i).

Calculating (ii):

Multiplying (equation 1) both sides by $\sin\left(\frac{m\pi x}{l}\right)$ and integrating from $-l$ to l , we have

$$\int_{-l}^l f(x) \sin\left(\frac{m\pi x}{l}\right) dx = \int_{-l}^l A \sin\left(\frac{m\pi x}{l}\right) dx + \sum_{n=1}^{\infty} \left\{ a_n \int_{-l}^l \sin\left(\frac{m\pi x}{l}\right) \cos\left(\frac{n\pi x}{l}\right) dx + b_n \int_{-l}^l \sin\left(\frac{m\pi x}{l}\right) \sin\left(\frac{n\pi x}{l}\right) dx \right\} \dots (3)$$

$\downarrow$
(a)

$\downarrow$
(b)

$\downarrow$
(c)

Now we'll calculate the values for above functions.

$$(a); \quad \int_{-l}^l A \sin\left(\frac{m\pi x}{l}\right) dx = 0 \quad \because \sin \alpha \text{ is odd } \forall \alpha \in R$$

$$(b); \quad \int_{-l}^l \sin\left(\frac{m\pi x}{l}\right) \cos\left(\frac{n\pi x}{l}\right) dx = 0 \quad \because \sin \alpha \cos \beta \text{ are odd functions } \forall \alpha, \beta \in R$$

$$(c); \quad \int_{-l}^l \sin\left(\frac{m\pi x}{l}\right) \sin\left(\frac{n\pi x}{l}\right) dx = \frac{1}{2} \int_{-l}^l \left\{ \cos\frac{(m-n)\pi x}{l} - \cos\frac{(m+n)\pi x}{l} \right\} dx = 0 \quad ; \quad m \neq n$$

For $m = n$, we have

$$\int_{-l}^l \sin\left(\frac{m\pi x}{l}\right) \cos\left(\frac{n\pi x}{l}\right) dx = \frac{1}{2} \int_{-l}^l \sin^2\left(\frac{m\pi x}{l}\right) dx = \frac{1}{2} \int_{-l}^l \left\{ 1 - \cos 2\left(\frac{m\pi x}{l}\right) \right\} dx = \frac{1}{2} |x|_{-l}^l = l$$

By putting values in equation 3, we get

$$\int_{-l}^l f(x) \sin\left(\frac{m\pi x}{l}\right) dx = 0 + 0 + b_m l \Rightarrow b_n = \frac{1}{l} \int_{-l}^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

This is our required solution for (ii).

Calculating (iii):

Integrating equation 1 from $-l$ to l on both sides,

$$\int_{-l}^l f(x) dx = \int_{-l}^l A dx + \sum_{n=1}^{\infty} \left\{ a_n \int_{-l}^l \cos\left(\frac{n\pi x}{l}\right) dx + b_n \int_{-l}^l \sin\left(\frac{n\pi x}{l}\right) dx \right\}$$

$$\int_{-l}^l f(x) dx = 2Al \Rightarrow 2A = \frac{1}{l} \int_{-l}^l f(x) dx \quad \dots (a)$$

Put $n = 0$ in a_n ;

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx \Rightarrow a_0 = \frac{1}{l} \int_{-l}^l f(x) dx \quad \dots (b)$$

Comparing a and b , we have $a_0 = 2A \Rightarrow A = \frac{a_0}{2}$

Hence equation 1 (Fourier Series) becomes

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right) \right]$$

LECTURE NO. 22

Problem:

Expand $f(x) = x^2$; $0 < x < 2\pi$ in a Fourier Series.

Solution: Fourier Series

$$f(x) = A + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right) \right]$$

According to given function, ($\because l = \pi$)

$$f(x) = A + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)] \quad \dots (A)$$

As we know from previous lectures,

$$a_n = \frac{1}{l} \int_c^{c+2l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

According to given function,

$$a_n = \frac{1}{\pi} \int_0^{2\pi} x^2 \cos\left(\frac{n\pi x}{\pi}\right) dx \Rightarrow \frac{1}{\pi} \int_0^{2\pi} x^2 \cos(nx) dx \quad \because l = \pi$$

By integrating and applying limits, we have

$$a_n = \frac{1}{\pi} \left[x^2 \left(\frac{\sin(nx)}{n} \right) - (2x) \left(-\frac{\cos(nx)}{n^2} \right) + (2) \left(-\frac{\sin(nx)}{n^3} \right) \right]_0^{2\pi}$$

$$a_n = \frac{1}{\pi} \left[4\pi^2(0) + \frac{4\pi(1)}{n^2} + 0 \right] = \frac{1}{\pi} \left[\frac{4\pi}{n^2} \right] = \frac{4}{n^2} \quad \dots (1)$$

Now

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

According to given function,

$$b_n = \frac{1}{\pi} \int_0^{2\pi} x^2 \sin(nx) dx$$

By integrating and applying limits, we have

$$b_n = \frac{1}{\pi} \left[x^2 \left(-\frac{\cos(nx)}{n} \right) - (2x) \left(-\frac{\sin(nx)}{n^2} \right) + 2 \left(\frac{\cos(nx)}{n^3} \right) \right]_0^{2\pi}$$

$$b_n = \frac{1}{\pi} \left[0 + \frac{4\pi^2(-1)}{n} + 0 \right] = -\frac{4\pi}{n} \quad \dots (2)$$

Now

$$A = \frac{a_0}{2} \Rightarrow \frac{1}{\pi} \int_0^{2\pi} f(x) dx \Rightarrow \frac{1}{\pi} \int_0^{2\pi} x^2 dx \Rightarrow \frac{1}{\pi} \left[\frac{x^3}{3} \right]_0^{2\pi} \Rightarrow \frac{8\pi^2}{3} \quad \dots (3)$$

Putting values in Fourier series,

$$f(x) = \frac{4\pi^2}{3} + \sum_{n=1}^{\infty} \left[\frac{4}{n^2} \cos(nx) - \frac{4\pi}{n} \sin(nx) \right]$$

LECTURE NO. 23

Fourier Expansion of Even Functions:

Theorem: Show that an even function does not have *sine* terms in its Fourier Series.

Proof: Since

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right) \right] \quad \dots (1)$$

If $f(x)$ is even function, then we know

$$f(x) = f(-x)$$

So, replacing x by $-x$, we have

$$f(-x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{-n\pi x}{l}\right) + b_n \sin\left(\frac{-n\pi x}{l}\right) \right]$$

As $\cos(-x) = \cos x$ & $\sin(-x) = -\sin x$, hence

$$f(-x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{l}\right) - b_n \sin\left(\frac{n\pi x}{l}\right) \right] \dots (2)$$

By comparing equation 1 & 2,

$$\begin{aligned} f(x) &= f(-x) \\ \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right) \right] &= \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{l}\right) - b_n \sin\left(\frac{n\pi x}{l}\right) \right] \\ \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) - \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) \end{aligned}$$

By cancelling the terms, we have

$$\begin{aligned} \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) &= - \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) \\ 2 \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) &= 0 \Rightarrow \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) = 0 \end{aligned}$$

So equation (1) implies,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right)$$

Hence proved that an even function does not have *sine* terms in its Fourier Series.

LECTURE NO. 24

Half Range Expansion of Identity Function:

Expand $f(x) = x$; $0 < x < 2$ in half-range *sines* & *cosine series*.

Solution:

(i): For odd extension of $f(x) = x$ in $(-2, 2)$ $\therefore -2 < x < 2$

This implies $2l = 4$ so $l = 2$

For odd expansion; $a_n = 0$

($\therefore a_0 = 0$ also) And

$$b_n = \frac{2}{l} \int_0^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

For given function,

$$b_n = \frac{2}{2} \int_0^2 x \sin\left(\frac{n\pi x}{2}\right) dx \Rightarrow \int_0^2 x \sin\left(\frac{n\pi x}{2}\right) dx$$

By integrating and applying limits, we have

$$\begin{aligned} b_n &= \left[x \left\{ \frac{-2}{n\pi} \cos\left(\frac{n\pi x}{2}\right) \right\} - (1) \left\{ \frac{4}{n^2\pi^2} \sin\left(\frac{n\pi x}{2}\right) \right\} \right]_0^2 \\ b_n &= -\frac{4}{n\pi} \cos(n\pi) \end{aligned}$$

Now, putting values in Fourier Series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right) \right]$$

$$f(x) = 0 + 0 + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{2}\right) \Rightarrow \sum_{n=1}^{\infty} \left(-\frac{4}{n\pi} \cos(n\pi) \right) \sin\left(\frac{n\pi x}{2}\right)$$

$$f(x) = \frac{4}{\pi} \left[\sin\left(\frac{\pi x}{2}\right) - \frac{1}{2} \sin\left(\frac{2\pi x}{2}\right) + \frac{1}{3} \sin\left(\frac{3\pi x}{2}\right) - \dots \right]$$

Required result.

LECTURE NO. 25

(ii): For even extension of $f(x) = |x|$ in $(-2, 2)$

This implies $2l = 4$ so $l = 2$

For odd expansion; $b_n = 0$ and

$$a_n = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

For given function,

$$a_n = \frac{2}{2} \int_0^2 x \cos\left(\frac{n\pi x}{2}\right) dx \Rightarrow \int_0^2 x \cos\left(\frac{n\pi x}{2}\right) dx$$

By integrating and applying limits, we have

$$a_n = \left[x \left\{ \frac{2}{n\pi} \sin\left(\frac{n\pi x}{2}\right) \right\} - (1) \left\{ \frac{-4}{n^2 \pi^2} \cos\left(\frac{n\pi x}{2}\right) \right\} \right]_0^2$$

$$a_n = 0 + \frac{4}{n^2 \pi^2} (\cos n\pi - 1) \Rightarrow \frac{4}{n^2 \pi^2} [(-1)^n - 1]$$

For $n = 0$,

$$a_0 = \frac{2}{l} \int_0^l x dx \Rightarrow \frac{2}{2} \int_0^2 x dx \Rightarrow \left[\frac{x^2}{2} \right]_0^2 \Rightarrow 2$$

$$\frac{a_0}{2} = 1$$

Now putting values in Fourier Series,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right) \right]$$

$$f(x) = 1 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{2}\right) + 0 \Rightarrow 1 + \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} [(-1)^n - 1] \cos\left(\frac{n\pi x}{2}\right)$$

$$f(x) = 1 + \frac{4}{\pi^2} \left[-2 \cos\left(\frac{\pi x}{2}\right) - \frac{2}{3^2} \cos\left(\frac{3\pi x}{2}\right) - \frac{2}{5^2} \cos\left(\frac{5\pi x}{2}\right) - \dots \right]$$

$$f(x) = 1 - \frac{8}{\pi^2} \left[\cos\left(\frac{\pi x}{2}\right) + \frac{1}{3^2} \cos\left(\frac{3\pi x}{2}\right) + \frac{1}{5^2} \cos\left(\frac{5\pi x}{2}\right) + \dots \right]$$

Required result.

LECTURE NO. 26

Parseval's Identity:

Theorem: If Fourier Series of $f(x)$ converges uniformly to $f(x)$ in $(-l, l)$, then prove that

$$\frac{1}{l} \int_{-l}^l \{f(x)\}^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$$

Proof:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right) \right] \dots (1)$$

Multiplying both sides by $f(x)$ and integrating from $-l$ to l ,

$$\int_{-l}^l \{f(x)\}^2 dx = \int_{-l}^l \frac{a_0}{2} f(x) dx + \sum_{n=1}^{\infty} \left[a_n \int_{-l}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx + b_n \int_{-l}^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx \right] \dots (A)$$

$\therefore$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx \Rightarrow a_n l = \int_{-l}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

Similarly for

$$b_n l = \int_{-l}^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

$$a_0 l = \int_{-l}^l f(x) dx$$

So, equation (A) can be written as

$$\int_{-l}^l \{f(x)\}^2 dx = \frac{a_0}{2} (a_0 l) + \sum_{n=1}^{\infty} [a_n (a_n l) + b_n (b_n l)]$$

$$\int_{-l}^l \{f(x)\}^2 dx = \frac{a_0^2}{2} (l) + (l) \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$$

$$\frac{1}{l} \int_{-l}^l \{f(x)\}^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$$

Required result proved.

LECTURE NO. 27

Problem:

Use Parseval's identity for $f(x) = |x|$, $-2 < x < 2$ to show that

$$\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots = \frac{\pi^4}{96}$$

Where it is given

$$a_0 = 2, \quad a_n = \frac{4}{n^2 \pi^2} \{(-1)^n - 1\}, n \neq 0 \quad \& \quad b_n = 0$$

Also deduce that

$$\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots = \frac{\pi^4}{90}$$

Solution: Parseval's identity

$$\frac{1}{l} \int_{-l}^l \{f(x)\}^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \dots (1)$$

For $f(x) = |x|$; $(-2, 2) \Rightarrow 2l = 4$ so $l = 2$

Equation (1) becomes

$$\frac{1}{2} \int_{-2}^2 x^2 dx = \frac{2^2}{2} + \sum_{n=1}^{\infty} \frac{16}{n^4 \pi^4} \{(-1)^n - 1\}^2 + 0$$

By solving, we have

$$\frac{8}{3} = 2 + \frac{64}{\pi^4} \left(\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots \right)$$

$$\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots = \frac{\pi^4}{96}$$

Proved. Now let

$$S = \frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots$$

$$S = \left(\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots \right) + \left(\frac{1}{2^4} + \frac{1}{4^4} + \frac{1}{6^4} + \dots \right)$$

$$S = \left(\frac{\pi^4}{96} \right) + \frac{1}{2^4} \left(\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots \right)$$

$$S = \left(\frac{\pi^4}{96} \right) + \frac{1}{2^4} (S)$$

$$S = \frac{\pi^4}{90}$$

Deduced.

LECTURE NO. 28

Finding a Fourier Series by Integration:

Theorem: Fourier series of $f(x)$ can be integrated by term from a to x and the resulting series will converge uniformly to $\int_0^x f(x)$ provided that;

(i): $f(x)$ is piecewise continuous in $(-l, l)$

(ii): $a, x \in (-l, l)$

Problem: Find the Fourier Series of $f(x) = x^2, 0 < x < 2$, by integrating

$$x = \frac{4}{\pi} \left[\sin \frac{\pi x}{2} - \frac{1}{2} \sin \frac{2\pi x}{2} + \frac{1}{3} \sin \frac{3\pi x}{2} - \dots \right]$$

And further evaluate;

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2} = ?$$

Solution: Given that

$$x = \frac{4}{\pi} \left[\sin \frac{\pi x}{2} - \frac{1}{2} \sin \frac{2\pi x}{2} + \frac{1}{3} \sin \frac{3\pi x}{2} - \dots \right]$$

Integrating from 0 to x term by term

$$\int_0^x x dx = \frac{4}{\pi} \left[\int_0^x \sin \frac{\pi x}{2} dx - \frac{1}{2} \int_0^x \sin \frac{2\pi x}{2} dx + \frac{1}{3} \int_0^x \sin \frac{3\pi x}{2} dx \right]$$

$$\left[\frac{x^2}{2} \right]_0^x = \frac{4}{\pi} \left[\frac{2}{\pi} \left| -\cos \frac{\pi x}{2} \right|_0^x + \frac{2}{\pi} \frac{1}{2^2} \left| \cos \frac{2\pi x}{2} \right|_0^x - \frac{2}{\pi} \frac{1}{3^2} \left| \cos \frac{3\pi x}{2} \right|_0^x - \dots \right]$$

$$\frac{x^2}{2} = \frac{4}{\pi} \left[\frac{2}{\pi} \left(-\cos \frac{\pi x}{2} + 1 \right) + \frac{2}{\pi} \frac{1}{2^2} \left(\cos \frac{2\pi x}{2} - 1 \right) - \frac{2}{\pi} \frac{1}{3^2} \left(\cos \frac{3\pi x}{2} - 1 \right) \right]$$

$$x^2 = \frac{16}{\pi^2} \left[1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots \right] - \frac{16}{\pi^2} \left[\cos \frac{\pi x}{2} - \frac{1}{2^2} \cos \frac{2\pi x}{2} + \frac{1}{3^2} \cos \frac{3\pi x}{2} - \dots \right]$$

Comparing above relation with general form;

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos \left(\frac{n\pi x}{l} \right) + b_n \sin \left(\frac{n\pi x}{l} \right) \right]$$

We have

$$\frac{a_0}{2} \Rightarrow \frac{1}{l} \int_0^l f(x) dx = \frac{16}{\pi^2} \left[1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots \right] \dots (1)$$

Here

$$\frac{1}{l} \int_0^l f(x) dx = \frac{1}{2} \int_0^2 x^2 dx = \frac{1}{2} \left[\frac{x^3}{3} \right]_0^2 = \frac{4}{3}$$

So equation (1) becomes,

$$\begin{aligned} \frac{4}{3} &= \frac{16}{\pi^2} \left[1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots \right] \\ \frac{4}{3} &= \frac{16}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2} \\ \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2} &= \frac{\pi^2}{12} \end{aligned}$$

Required result.

LECTURE NO. 29

Problem:

Check the term by term differentiation of Fourier Series;

$$f(x) = \frac{4}{\pi} \left[\sin \frac{\pi x}{2} - \frac{1}{2} \sin \frac{2\pi x}{2} + \frac{1}{3} \sin \frac{3\pi x}{2} - \dots \right]$$

Solution: Taking derivative,

$$\begin{aligned} f'(x) &= \frac{4}{\pi} \left[\frac{\pi}{2} \cos \frac{\pi x}{2} - \frac{1}{2} \frac{2\pi}{2} \cos \frac{2\pi x}{2} + \frac{1}{3} \frac{3\pi}{2} \cos \frac{3\pi x}{2} - \dots \right] \\ f'(x) &= 2 \left[\cos \frac{\pi x}{2} - \cos \frac{2\pi x}{2} + \cos \frac{3\pi x}{2} - \dots \right] \end{aligned}$$

This implies

$$a_n = (-1)^{n-1} \left(\cos \frac{n\pi x}{2} \right) \neq 0$$

$\Rightarrow$ Series does not converge $\Rightarrow$ It does not converge uniformly $\Rightarrow$ Term by term differentiation is not possible.

LECTURE NO. 30

Heat Flow Problem:

A bar of length "l" whose entire surface is insulated including its ends at $x = 0$ and $x = l$. Its initial temperature is $f(x)$, then determine the subsequent temperature of the bar.

Solution: It is a heat flow boundary value problem (B.V.P.). As we know heat equation

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2} \dots (1)$$

Where

$$|u(x, t)| < M, \quad u(x, 0) = f(x), \quad u_x(0, t) = 0 = u_x(l, t)$$

Let $u(x, t) = X(x)T(t) = XT$ be its solution. Equation (1) implies,

$$\begin{aligned} XT' &= kX''T \\ \frac{T'}{kT} &= \frac{X''}{X} = -\lambda^2 \text{ (say)} \end{aligned}$$

Here become two equations. We'll solve them side by side.

$$T' + \lambda^2 kT = 0 \dots (i)$$

$$\text{Let } T = e^{mt}$$

$$T' = me^{mt} \Rightarrow mT$$

$$\text{So (i); } mT + \lambda^2 kT = 0$$

$$m = -\lambda^2 k$$

$$\therefore T = ce^{-\lambda^2 kt}$$

$$X'' + \lambda^2 X = 0 \dots (ii)$$

$$X = e^{mx} \Rightarrow X'' = m^2 e^{mx} = m^2 X$$

$$m^2 X + \lambda^2 X = 0 \Rightarrow m = \pm i\lambda \quad (\because x \neq 0)$$

$$X = ae^{\lambda ix} + be^{-\lambda ix}$$

$$X = a(\cos \lambda x + i \sin \lambda x) + b(\cos \lambda x - i \sin \lambda x)$$

$$X = (a + b) \cos \lambda x + i(a - b) \sin \lambda x = A \cos \lambda x + B \sin \lambda x$$

By putting values,

$u(x, t) = X(x)T(t) = XT = ce^{-\lambda^2 kt} [A\cos\lambda x + B\sin\lambda x] = e^{-\lambda^2 kt} [\alpha\cos\lambda x + \beta\sin\lambda x] \dots (2)$

Now

$$u_x = e^{-\lambda^2 kt} [-\alpha\lambda \cos\lambda x + \beta\lambda \sin\lambda x]$$
$$u_x(0, t) = \beta \lambda e^{-\lambda^2 kt} = 0 \Rightarrow \beta = 0$$

Equation (2) implies,

$$u(x, t) = \alpha e^{-\lambda^2 kt} \cos(\lambda x) \dots (3)$$
$$u_x = -\lambda \alpha e^{-\lambda^2 kt} \sin(\lambda x)$$
$$u_x(l, t) = -\lambda \alpha e^{-\lambda^2 kt} \sin(\lambda l) = 0$$
$$\Rightarrow \sin(\lambda l) = 0 \Rightarrow \lambda l = m\pi \Rightarrow \lambda = \frac{m\pi}{l}$$

Equation (3) implies,

$$u(x, t) = \alpha e^{-\frac{m^2 \pi^2}{l^2} kt} \cos\left(\frac{m\pi}{l} x\right)$$

By using super-position principle;

$$u(x, t) = \frac{a_0}{2} + \sum_{m=1}^{\infty} \alpha_m e^{-\frac{m^2 \pi^2}{l^2} kt} \cos\left(\frac{m\pi}{l} x\right) \dots (A)$$

Now; as given

$$u(x, 0) = f(x) = \frac{a_0}{2} + \sum_{m=1}^{\infty} \alpha_m \cos\left(\frac{m\pi}{l} x\right)$$

Where

$$\alpha_m = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{m\pi}{l} x\right) dx, \quad \frac{a_0}{2} = \frac{1}{l} \int_0^l f(x) dx$$

So, equation (A) becomes (by putting values)

$$u(x, t) = \frac{1}{l} \int_0^l f(x) dx + \frac{2}{l} \sum_{m=1}^{\infty} \left\{ \int_0^l f(x) \cos\left(\frac{m\pi}{l} x\right) e^{-\frac{m^2 \pi^2}{l^2} kt} \cos\left(\frac{m\pi}{l} x\right) dx \right\}$$

Required result.

LECTURE NO. 31

Laplace Equation:

Problem: Suppose that the three sides of a square plate kept at zero temperature and fourth one at u_1 . Determine the steady state temperature at all parts on the plate.

For Solution; [WATCH](#) Lecture (Lengthy Calculations)

LECTURE NO. 32

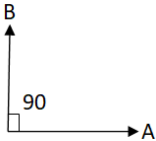
Orthogonal Functions And Orthogonal Sets:

- (i): Can we generalize the idea of vectors and orthogonality?
- $\{f(x) \text{ defined on } (a, b); \text{ value of 'f' at each point on } (a, b) \text{ represents its comoponents.}$
- (ii): Say $A(x)$ & $B(x)$ defined on (a, b) if $\int_a^b A(x).B(x) = 0$, then $A(x)$ & $B(x)$ are said to orthogonal.
- (iii): A vector say $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ is said to normalized if its magnitude is unity i.e.

$$\hat{r} = \frac{\vec{r}}{|\vec{r}|} = \frac{x\hat{i} + y\hat{j} + z\hat{k}}{\sqrt{x^2 + y^2 + z^2}}$$

(iv): $A(x)$ normal or normalized in (a, b) if

$$\int_a^b \{A(x)\}^2 dx = 1$$



Orthogonal Sets:

A set of function say $\{\Phi_k(x)\}, k = 1,2,3, \dots$ defined on (a, b) such that

(i):

$$\int_a^b \Phi_m(x) \cdot \Phi_n(x) = 0 \quad , \quad m \neq n$$

(ii):

$$\int_a^b \{\Phi_m(x)\}^2 = 1 \quad , \quad m = 1,2,3, \dots$$

Then the set $\{\Phi_k(x)\}, k = 1,2,3, \dots$ is orthogonal.

(i)& (ii) implies,

$$\int_a^b \Phi_m(x) \cdot \Phi_n(x) dx = S_{mn} = \begin{cases} 0 & \text{if } m \neq n \\ 1 & \text{if } m = n \end{cases}$$

Orthogonality w.r.to Weight Function:

If

$$\int_a^b \Phi_m(x) \Phi_n(x) \omega(x) dx = S_{mn}$$

Where $\omega(x) \geq 0$, then $\{\Phi_m(x)\}_{k=1}^\infty$ is orthogonal w.r.to weight function.

LECTURE NO. 33

Obtaining Normalizing Constants From orthogonal Sets:

Show that the set

$$\left\{1, \sin \frac{\pi x}{l}, \cos \frac{\pi x}{l}, \sin \frac{2\pi x}{l}, \cos \frac{2\pi x}{l}, \dots \right\}$$

Is an orthogonal set. Also find its corresponding normalizing constants, so that given set is orthogonal.

Solution: As we know condition for orthogonal set condition

$$\int_{-l}^l \Phi_m(x) \cdot \Phi_n(x) dx = \begin{cases} 0 & \text{if } m \neq n \\ 1 & \text{if } m = n \end{cases}$$

Possibilities for given set (using Fourier series results;

(i):

$$\int_{-l}^l 1 \sin \frac{k\pi x}{l} dx = 0 = \int_{-l}^l 1 \cos \frac{k\pi x}{l} dx \quad \forall k = 1,2,3, \dots$$

(ii):

$$\int_{-l}^l \sin \frac{k\pi x}{l} \cos \frac{p\pi x}{l} dx = 0 \quad k \neq p$$

(iii):

$$\int_{-l}^l \sin \frac{m\pi x}{l} \sin \frac{n\pi x}{l} dx = \int_{-l}^l \cos \frac{m\pi x}{l} \cos \frac{n\pi x}{l} dx = \begin{cases} 0 & \text{if } m \neq n \\ l & \text{if } m = n \end{cases}$$

(iii) Implies

$$\int_{-l}^l \sin^2 \frac{m\pi x}{l} dx = l$$

$$\int_{-l}^l \cos^2 \frac{m\pi x}{l} dx = l$$

$$\int_{-l}^l \left(\frac{1}{\sqrt{l}} \sin \frac{m\pi x}{l}\right)^2 dx = 1$$

$$\int_{-l}^l \left(\frac{1}{\sqrt{l}} \cos \frac{m\pi x}{l}\right)^2 dx = 1$$

Similarly for (i):

$$\int_{-l}^l (1)^2 dx = 2l \Rightarrow \int_{-l}^l \left(\frac{1}{\sqrt{2l}}\right)^2 dx = 1$$

Hence, Orthonormalizing constants;

$$\frac{1}{\sqrt{2l}}, \frac{1}{\sqrt{l}}$$

Corresponding Orthonormal set;

$$\left\{ \frac{1}{\sqrt{2l}}, \frac{1}{\sqrt{l}} \sin \frac{\pi x}{l}, \frac{1}{\sqrt{l}} \cos \frac{\pi x}{l}, \frac{1}{\sqrt{l}} \sin \frac{2\pi x}{l}, \frac{1}{\sqrt{l}} \cos \frac{2\pi x}{l}, \dots \right\}$$

LECTURE NO. 34

Generalized Fourier Series:

Given that $\{\Phi_m(x)\}_{m=1}^\infty$ be an orthogonal set of functions and if possible to expand a function " $f(x)$ " in a set of orthonormal functions, i.e.

$$f(x) = \sum_{n=1}^\infty C_n \Phi_n(x) \quad ; \quad a \leq x \leq b$$

Then such series is called Orthonormal Series or Generalized Fourier Series and " C_n " are called Generalized Fourier Coefficients.

If $f(x)$ & $f'(x)$ are piecewise continuous functions, then

$$\sum_{n=1}^\infty C_n \Phi_n(x) = \frac{1}{2} [f(x+0) + f(x-0)]$$

“If a function and its derivative are piecewise continuous, then the series converge to their average”.

LECTURE NO. 35

Theorem:

Let $\{\Phi_n(x)\}$ be a set of mutually orthonormal functions in (a, b) . Show that if $\sum_{n=1}^\infty C_n \Phi_n(x)$ converges uniformly, then $C_n = \int_a^b f(x) \Phi_n(x) dx$.

Proof: As given that

The series $\sum_{n=1}^\infty C_n \Phi_n(x)$ converges uniformly to $f(x)$

This implies,

$$f(x) = \sum_{n=1}^\infty C_n \Phi_n(x) \quad \dots (1)$$

Multiplying both sides by $\Phi_m(x)$ and integrating from ' a to b '

$$\int_a^b f(x) \Phi_m(x) dx = \sum_{n=1}^\infty C_n \int_a^b \Phi_m(x) \Phi_n(x) dx \quad \dots (2)$$

As

$$\int_a^b \Phi_m(x) \Phi_n(x) dx = \begin{cases} 0 & \text{if } m \neq n \\ l & \text{if } m = n \end{cases}$$

Hence equation (2) implies

$$C_m = \int_a^b f(x) \Phi_m(x) dx$$

Or it can be written as;

$$C_n = \int_a^b f(x) \Phi_n(x) dx$$

Which are called the generalized fourier constants.

LECTURE NO. 36

MSE and RMSE:

MSE stands for "Mean Square Error" and RMSE stands for "Root Mean Square Error".

Let $f(x)$ and $f'(x)$ be continuous piecewise functions in (a, b) , $\{\Phi_m(x)\}_{m=1}^{\infty}$ be an orthogonal set in (a, b) .

Now suppose the sum $S_m(x) = \sum_{n=1}^{\infty} \alpha_n \Phi_n(x)$ be an approximation of $f(x)$, where " α_n " are unknown constants. Then the mean square error of this approximation is given by,

$$MSE = \frac{\int_a^b \{f(x) - S_m(x)\}^2 dx}{b - a}$$

And

$$RMSE = \sqrt{\frac{\int_a^b \{f(x) - S_m(x)\}^2 dx}{b - a}}$$

These are used to compare the accuracy of different mathematical methods of a physical system.

We are aimed to find " α_n " which will produce the least mean square error.

Use: (MSE; for large or greater errors/quantities & RMSE; for same units as that of given function)

LECTURE NO. 37

Least Square Approximation & Principle of Finality:

Theorem: The root mean square error (RMSE) is least when the coefficients " α_n " are equal to generalized Fourier Coefficient; i.e.

$$\alpha_n = c_n = \frac{1}{b-a} \int_a^b f(x) \Phi_n(x) dx$$

Proof: As we known

$$RMSE = \sqrt{\frac{\int_a^b \{f(x) - S_m(x)\}^2 dx}{b - a}} \Rightarrow \frac{1}{\sqrt{b-a}} \sqrt{\int_a^b \{f(x) - S_m(x)\}^2 dx}$$

Now

$$f(x) - S_m(x) = f(x) - \sum_{n=1}^{\infty} \alpha_n \Phi_n(x)$$

Where $\{\Phi_n(x)\}_{n=1}^{\infty}$ is an orthonormal set and " α_n " is unknown. Now,

$$\begin{aligned} \{f(x) - S_m(x)\}^2 &= \{f(x)\}^2 - 2 \sum_{n=1}^{\infty} \alpha_n \Phi_n(x) f(x) + \left\{ \sum_{n=1}^{\infty} \alpha_n \Phi_n(x) \right\}^2 \quad \dots (1) \\ \left\{ \sum_{n=1}^{\infty} \alpha_n \Phi_n(x) \right\}^2 &= \sum_{n=1}^{\infty} \alpha_n \Phi_n(x) \times \sum_{n=1}^{\infty} \alpha_n \Phi_n(x) = \sum_{n=1}^{\infty} \alpha_n \Phi_n(x) \times \sum_{m=1}^{\infty} \alpha_m \Phi_m(x) \\ \left\{ \sum_{n=1}^{\infty} \alpha_n \Phi_n(x) \right\}^2 &= \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \alpha_n \alpha_m \Phi_n(x) \Phi_m(x) \end{aligned}$$

By putting equation (1) becomes,

$$\{f(x) - S_m(x)\}^2 = \{f(x)\}^2 - 2 \sum_{n=1}^{\infty} \alpha_n \Phi_n(x) f(x) + \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \alpha_n \alpha_m \Phi_n(x) \Phi_m(x)$$

Integrating from "a to b"

$$\int_a^b \{f(x) - S_m(x)\}^2 dx = \int_a^b \{f(x)\}^2 dx - 2 \sum_{n=1}^{\infty} \alpha_n \int_a^b \Phi_n(x) f(x) dx + \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \alpha_n \alpha_m \int_a^b \Phi_n \Phi_m dx \quad \dots (2)$$

$$\therefore \int_a^b \Phi_n(x) f(x) dx = c_n = \text{Generalized Fourier Coefficients}$$

So, equation (2) becomes,

$$\begin{aligned} \int_a^b \{f(x) - S_m(x)\}^2 dx &= \int_a^b \{f(x)\}^2 dx - 2 \sum_{n=1}^{\infty} \alpha_n c_n + \sum_{n=1}^{\infty} (\alpha_n)^2 \\ \int_a^b \{f(x) - S_m(x)\}^2 dx &= \int_a^b \{f(x)\}^2 dx + \sum_{n=1}^{\infty} (\alpha_n^2 - 2 \alpha_n c_n) \dots (3) \\ \{\alpha_n^2 - 2 \alpha_n c_n\} &= \alpha_n^2 - 2 \alpha_n c_n + (c_n)^2 - (c_n)^2 = (\alpha_n - c_n)^2 - (c_n)^2 \end{aligned}$$

So, equation (3) becomes,

$$\begin{aligned} \int_a^b \{f(x) - S_m(x)\}^2 dx &= \int_a^b \{f(x)\}^2 dx + \sum_{n=1}^{\infty} \{(\alpha_n - c_n)^2 - (c_n)^2\} \\ \int_a^b \{f(x) - S_m(x)\}^2 dx &= \int_a^b \{f(x)\}^2 dx + \sum_{n=1}^{\infty} (\alpha_n - c_n)^2 - \sum_{n=1}^{\infty} (c_n)^2 \end{aligned}$$

Error will be minimum when

$$\left\{ \sum_{n=1}^{\infty} (\alpha_n - c_n)^2 = 0 \Rightarrow (\alpha_n - c_n)^2 = 0 \Rightarrow \alpha_n - c_n \Rightarrow \alpha_n = c_n \right.$$

LECTURE NO. 38

Bissel's Inequality:

Theorem: For generalized Fourier coefficients " c_n " corresponding to $f(x)$, show that

$$\sum_{n=1}^{\infty} (c_n)^2 \leq \int_a^b \{f(x)\}^2 dx$$

Proof: Since $RMSE \geq 0$, so

$$\sqrt{\frac{\int_a^b \{f(x) + S_m(x)\}^2 dx}{b-a}} \geq 0 \Rightarrow \sqrt{\int_a^b \{f(x) + S_m(x)\}^2 dx} \geq 0$$

$$\Rightarrow \int_a^b \{f(x) + S_m(x)\}^2 dx \geq 0$$

$$\Rightarrow \int_a^b \{f(x)\}^2 dx - \sum_{n=1}^{\infty} (c_n)^2 \geq 0$$

$$\Rightarrow \sum_{n=1}^{\infty} (c_n)^2 \leq \int_a^b \{f(x)\}^2 dx$$

This is our required result. This inequality is known as Bissel's inequality.

LECTURE NO. 39

Limiting Value of Generalized Fourier Coefficients:

Theorem: Show that

$$\lim_{n \rightarrow \infty} \int_a^b f(x) \Phi_n(x) dx = 0$$

Proof: As we know, Generalized Fourier Coefficients

$$c_n = \int_a^b f(x)\Phi_n(x) \, dx$$

And from Bissel’s inequality

$$\sum_{n=1}^\infty (c_n)^2 \leq \int_a^b f(x)\Phi_n(x) \, dx$$

Since

$$\int_a^b f(x)\Phi_n(x) \, dx < k \quad \forall k \in R$$

However large it may be.

$$\sum_{n=1}^\infty (c_n)^2 \text{ will converge} \Rightarrow c_n \rightarrow 0 \text{ as } n \rightarrow \infty$$

This implies that

$$\lim_{n \rightarrow \infty} c_n = 0 \Rightarrow \lim_{n \rightarrow \infty} \int_a^b f(x)\Phi_n(x) \, dx = 0$$

This is our required reslut.

LECTURE NO. 40

Sturm-Liouville System:

A boundary value problem of the form:

$$\frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] + [q(x) + \lambda r(x)]y = 0 \quad a \leq x \leq b \quad \dots (1)$$

Subjected to

$$\alpha_1 y(a) + \alpha_2 y'(a) = 0 \quad , \quad \beta_1 y(b) + \beta_2 y'(b) = 0 \quad \dots (2)$$

Where $\alpha_1, \alpha_2, \beta_1, \beta_2$ are given constants and $p(x), q(x)$ and $r(x)$ are given differentiable functions. And λ is unspecified parameter independent of x .

Here system has a squence of eigon value λ_n and corresponding eigen functions satisfying boundary value problem.

Example: $y'' + \lambda y = 0 \quad ; \quad B.V.P: y(0) = y(1) = 0$

This implies

$$\frac{d}{dx} \left[1 \cdot \frac{dy}{dx} \right] + [(0 + \lambda \cdot 1)y] = 0 \quad 0 \leq x \leq 1$$

Boundary values can be written as,

$$\begin{aligned} y(0) &= 1 y(0) + 0 y'(0) = 0 = \alpha_1 y(a) + \alpha_2 y'(a) \\ y(1) &= 1 y(1) + 0 y'(1) = 0 = \beta_1 y(b) + \beta_2 y'(b) \end{aligned}$$

Comparing this with Sturm-Liouville System, we have

$$\alpha_1 = 1 \quad , \quad \alpha_2 = 0 \quad , \quad \beta_1 = 1 \quad , \quad \beta_2 = 0 \quad , \quad p(x) = 1 \quad , \quad q(x) = 0 \quad , \quad r(x) = 1$$

LECTURE NO. 41

Heat Equation as Motivation for S-L System:

Consider a B.V.P,

$$g(x) \frac{\partial u}{\partial t} = \frac{\partial}{\partial x} \left[k(x) \frac{\partial u}{\partial x} \right] + h(x) u \quad \dots (1)$$

Where

$$0 < x < l \quad , \quad t > 0 \quad , \quad u(0, t) = 0 \quad , \quad u(l, t) = 0 \quad , \quad u(x, 0) = f(x) \quad \& \quad |u(x, t)| < M$$

Let

$$u(x, t) = X(x).T(t) = XT$$

So, equation (1) can be written as

$$\begin{aligned} g(x) \, XT' &= \frac{\partial}{\partial x} [k(x)XT'] + h(x) \, XT \\ g(x) \, XT' &= T \frac{d}{dx} \left[k(x) \frac{dX}{dx} \right] + h(x) \, XT \end{aligned}$$

Dividing by $g(x)XT$, we get

$$\begin{aligned} \frac{T'}{T} &= \frac{1}{g(x)X} \frac{d}{dx} \left[k(x) \frac{dX}{dx} \right] + \frac{h(x)}{g(x)} = -\lambda \text{ (say)} \\ T' + \lambda T &= 0 \quad \dots (2) \end{aligned}$$

Boundary Values can be written as,

$$\begin{aligned} u(0,t) = 0 &\Rightarrow X(0).T(t) = 0 \Rightarrow X(0) = 0 \\ u(l,t) = 0 &\Rightarrow X(l).T(t) = 0 \Rightarrow X(l) = 0 \end{aligned}$$

On comparing;

$$y = x \quad , \quad p(x) = k(x) \quad , \quad q(x) = h(x) \quad , \quad r(x) = g(x)$$

Solution: From equation (2)

$$T' + \lambda T = 0 \Rightarrow T = C \, e^{-\lambda t}$$

By superposition principle

$$u(x,t) = \sum_{n=1}^{\infty} C_n \, e^{-\lambda n t} \, X_n(x) \quad \dots (3)$$

As given

$$\begin{aligned} u(x,0) &= f(x) \\ f(x) = u(x,0) &= \sum_{n=1}^{\infty} C_n \, X_n(x) \end{aligned}$$

If C_n is generalized coefficient, then

$$C_n = \int_0^l f(x) \, X_n(x) \, dx$$

Putting C_n value in equation (3) we have

$$u(x,t) = \sum_{n=1}^{\infty} \int_0^l f(x) \, X_n(x) \, dx \, e^{-\lambda n t} \, X_n(x)$$

This is our required result.

LECTURE NO. 42

Eigen Values and Eigen Functions of S-L System:

Given a S-L System;

$$\frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] + [q(x) + \lambda \, r(x)]y = 0$$

Subjected to;

$$\alpha_1 \, y(a) + \alpha_2 \, y'(a) = 0 \quad , \quad \beta_1 \, y(b) + \beta_2 \, y'(b) = 0 \quad a \leq x \leq b$$

Here non-trivial solution of S-L System exists in generalized for a particular value of λ . These values are called the Eigen values and the corresponding non-trivial solutions are called Eigen functions of S-L system.

Example:

$$y'' + \lambda y = 0 ; \quad y(0) = y(1) = 0 \quad , \quad 0 \leq x \leq 1$$

Let

$$y = e^{mx} \Rightarrow y'' = m^2 y \Rightarrow m = \pm i\sqrt{\lambda} ; \quad y \neq 0$$

By superposition principle;

$$y = A \cos\sqrt{\lambda}x + B \sin\sqrt{\lambda}x \quad \dots (1)$$

For $y(0) = A = 0$. So equation (1) becomes

$$y = B \sin\sqrt{\lambda}x$$

And for

$$y(1) = B \sin \sqrt{\lambda} = 0 \Rightarrow B \neq 0 \text{ \& so } \sin \sqrt{\lambda} = 0$$

This implies,

$$\sqrt{\lambda} = m\pi \Rightarrow \lambda_m = m^2\pi^2; \quad m \in Z$$

These are called Eigen values of given S-L System. And

$$y_m = B_m \sin m\pi x$$

These are called the Corresponding Eigen Functions.

LECTURE NO. 43

Orthogonality of Eigen Functions:

$$\begin{aligned} y_m &= B_m \sin m\pi x; \quad 0 \leq x \leq 1 \\ \int_0^1 (B_m \sin m\pi x)(B_n \sin n\pi x) \, dx \\ &= B_m B_n \int_0^1 (\sin m\pi x)(\sin n\pi x) \, dx \end{aligned}$$

Using relation $\{2 \sin A \sin B = \cos(A - B) - \cos(A + B)\}$, we have

$$= \frac{B_m B_n}{2} \int_0^1 \{\cos(m - n) \pi x - \cos(m + n) \pi x\} dx$$

By integrating and applying limits, we get

$$= \frac{B_m B_n}{2} \left[\frac{\sin(m - n) \pi x}{(m - n) \pi} - \frac{\sin(m + n) \pi x}{(m + n) \pi} \right]_0^1 = 0$$

Hence, $\{B_m \sin m\pi x\}_{m=1}^\infty$ is an orthogonal set.

LECTURE NO. 44

Normalization of Eigen Functions:

Given set $\{\Phi_m(x)\}_{m=1}^\infty$ is orthonormal if

$$\int_a^b \Phi_m \Phi_n \, dx = S_{mn} = \begin{cases} 0; & m \neq n \\ 1; & m = n \end{cases}$$

Now for $\{B_m \sin m\pi x\}_{m=1}^\infty$ in $(0,1)$, we apply the condition of orthonormality, if

$$\begin{aligned} \int_0^1 (B_m \sin m\pi x)^2 \, dx &= 1; \quad \text{iff } m = n \\ (B_m)^2 \int_0^1 \sin^2 m\pi x \, dx &= 1 \Rightarrow (B_m)^2 \int_0^1 \left(\frac{1 - \cos 2m\pi x}{2} \right) dx = 1 \\ (B_m)^2 \left[\frac{1}{2} x - \frac{\sin 2m\pi x}{42m\pi} \right]_0^1 &= 1 \Rightarrow (B_m)^2 \left[\frac{1}{2} (1) - 0 - 0 + 0 \right] = 1 \\ (B_m)^2 &= 2 \Rightarrow B_m = \sqrt{2}; \quad \forall m \in N \end{aligned}$$

Hence, $\{\sqrt{2} \sin m\pi x\}_{m=1}^\infty$ is an orthonormal set.

LECTURE NO. 45

Expansion in Terms of Orthonormal Functions:

Let $f(x) = 1$ (say) and we want to express $f(x)$ as an expansion of orthonormal functions say;

$$\{\Phi_n(x)\}_{n=1}^{\infty} = \{\sqrt{2} \sin n\pi x\}_{n=1}^{\infty}$$

i.e.

$$f(x) = \sum_{n=1}^{\infty} C_n \Phi_n(x) \quad \dots (1)$$

Where

$$C_n = \int_0^l f(x) \Phi_n(x) dx \Rightarrow \int_0^l 1 \cdot \sqrt{2} \sin n\pi x dx \Rightarrow \sqrt{2} \int_0^l \sin n\pi x dx$$

$$C_n = -\frac{\sqrt{2}}{n\pi} [\cos n\pi x]_0^l \Rightarrow -\frac{\sqrt{2}}{n\pi} [\cos n\pi - 1] \Rightarrow \frac{\sqrt{2}}{n\pi} [1 - (-1)^2]$$

So, equation (1) can be written as

$$f(x) = \sum_{n=1}^{\infty} \frac{\sqrt{2}}{n\pi} [1 - (-1)^2] \sin n\pi x$$

This is our required result.

LECTURE NO. 46

Characterizing the Eigen Values of S-L System:

Theorem: show that the Eigen values of a S-L System are real.

Proof: Given a S-L System in (a, b) ;

$$\frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] + [q(x) + \lambda r(x)] y = 0$$

Or it can be written as,

$$y \frac{d}{dx} (p y') + [q(x) + \lambda r(x)] y = 0 \quad \dots (1)$$

Subjected to boundary condition;

$$\alpha_1 y(a) + \alpha_2 y'(a) = 0 \quad \dots (a) \quad , \quad \beta_1 y(b) + \beta_2 y'(b) = 0 \quad \dots (b)$$

Where $\alpha_1, \alpha_2, \beta_1, \beta_2$ are given real numbers and $p(x), q(x)$ and $r(x)$ are real valued functions.

Taking conjugate of equation (1),

$$\bar{y} \frac{d}{dx} (p \bar{y}') + [q(x) + \bar{\lambda} r(x)] \bar{y} = 0 \quad \dots (2)$$

Subjected to;

$$\alpha_1 \bar{y}(a) + \alpha_2 \bar{y}'(a) = 0 \quad \dots (c) \quad , \quad \beta_1 \bar{y}(b) + \beta_2 \bar{y}'(b) = 0 \quad \dots (d)$$

By $y(2) - y'(1)$;

$$y \frac{d}{dx} [p \bar{y}'] - \bar{y} \frac{d}{dx} [p y'] + [q(x) + \bar{\lambda} r(x) - q(x) - \lambda r(x)] y \bar{y} = 0$$

$$y[p \bar{y}'' + p' \bar{y}'] - \bar{y}[p y'' + p' y'] + (\bar{\lambda} - \lambda) r(x) y \bar{y} = 0$$

$$p[y \bar{y}'' - \bar{y} y''] + p'[y \bar{y}' - \bar{y} y'] = (\lambda - \bar{\lambda}) r(x) y \bar{y}$$

$$p \frac{d}{dx} [y \bar{y}' - \bar{y} y'] + p'[y \bar{y}' - \bar{y} y'] = (\lambda - \bar{\lambda}) r(x) y \bar{y}$$

$$\frac{d}{dx} [p(x) \{y \bar{y}' - \bar{y} y'\}] = (\lambda - \bar{\lambda}) r(x) y \bar{y}$$

Integrating from "a to b",

$$(\lambda - \bar{\lambda}) \int_a^b r(x) y \bar{y} dx = p(x) [\{y \bar{y}' - \bar{y} y'\}]_{x=a}^{x=b}$$

$$= p(x) \{y(b) \bar{y}'(b) - \bar{y}(b) y'(b) - y(a) \bar{y}'(a) + \bar{y}(a) y'(a)\} \quad \dots (3)$$

Since,

$$\alpha_1 y(a) + \alpha_2 y'(a) = 0 \quad , \quad \beta_1 y(b) + \beta_2 y'(b) = 0$$

This implies,

$$\frac{y(a)}{y'(a)} = -\frac{\alpha_2}{\alpha_1} \quad , \quad \frac{\bar{y}(a)}{\bar{y}'(a)} = -\frac{\alpha_2}{\alpha_1}$$
$$\frac{y(a)}{y'(a)} = \frac{\bar{y}(a)}{\bar{y}'(a)} \Rightarrow y(a)\bar{y}'(a) - \bar{y}(a)y'(a) = 0$$

Similarly,

$$y(b)\bar{y}'(b) - \bar{y}(b)y'(b) = 0$$

So, equation (3) becomes, ($\therefore y\bar{y} = |y|^2$)

$$(\lambda - \bar{\lambda}) \int_a^b r(x)|y|^2 \, dx = 0$$

As,

$$r(x) \geq 0 \quad \text{and} \quad |y|^2 \geq 0 \Rightarrow \int_a^b r(x)|y|^2 \, dx \geq 0$$

So,

$$(\lambda - \bar{\lambda}) = 0 \Rightarrow \lambda = \bar{\lambda} \Rightarrow \lambda \text{ is real valued.}$$

Hence proved.

LECTURE NO. 47

The Gamma Function (Definition and Recurrence Relation):

The Gamma function is defined and given by;

$$\gamma(n+1) = \int_0^\infty x^n e^{-x} \, dx \quad \text{convergent for } n > 0.$$
$$= [x^n \cdot (e^{-x})]_0^\infty - \int_0^\infty (-e^{-x}) \cdot nx^{n-1} \, dx = 0 \Rightarrow 0 + n \int_0^\infty x^{n-1} \cdot e^{-x} \, dx$$
$$\gamma(n+1) = n \gamma(n) ; \quad \forall n > 0 \quad \dots (1)$$

If $n \in N$

For $n = 1$; $\gamma(2) = 1. \gamma(1) = 1.1 = 1!$

For $n = 2$; $\gamma(3) = 2. \gamma(2) = 2.1! = 2!$

For $n = 3$; $\gamma(4) = 3. \gamma(3) = 3.2! = 3!$

By induction;

$$\gamma(n+1) = n!$$

Example:

$$\gamma(6) = 5! = 5.4.3.2.1 = 120$$

LECTURE NO. 48

Analytic Continuation of Gamma Function:

The Gamma function is defined and given by;

$$\gamma(n+1) = \int_0^\infty x^n e^{-x} \, dx ; \quad n > 0.$$

It is technique which is used to extend the domain of a given function beyond its natural domain of definition.

$$\gamma(\alpha+1) = \alpha \gamma(\alpha) \Rightarrow \gamma(\alpha) = \frac{\gamma(\alpha+1)}{\alpha} \dots (a)$$

Here

$$\gamma(\alpha+1) = \frac{\gamma(\alpha+2)}{(\alpha+1)}$$

Hence above (a) relation becomes,

$$\gamma(\alpha) = \frac{1}{\alpha(\alpha+1)} \gamma(\alpha+2)$$

Similarly, we can write it as

$$\gamma(\alpha) = \frac{1}{\alpha(\alpha+1)(\alpha+2)} \gamma(\alpha+3)$$

In general form it can be written as,

$$\gamma(\alpha) = \frac{1}{\alpha(\alpha+1)(\alpha+2) \dots (\alpha+k)} \gamma(\alpha+k+1)$$

$\alpha > 0 \Rightarrow \gamma(\alpha)$ is not defined for $\alpha = 0, -1, -2, \dots$

Example:

$$\gamma(-0.5) = \frac{\gamma(-0.5+1)}{-0.5} = \frac{\gamma(0.5)}{-0.5} = \frac{\sqrt{\pi}}{-0.5} = -2\sqrt{\pi}$$

LECTURE NO. 49

An Important Gamma Value:

Theorem: Prove that $\gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.

Proof: Since,

$$\gamma(n+1) = \int_0^{\infty} x^{n-1} e^{-x} dx \Rightarrow \int_0^{\infty} x^{-1/2} e^{-x} dx$$

Let

$$\begin{cases} x = u^2 \Rightarrow dx = 2u du \\ x^{-\frac{1}{2}} = u^{-1} \Rightarrow u^{-1} = \frac{1}{u} \end{cases}$$

By putting, we get

$$\gamma\left(\frac{1}{2}\right) = \int_0^{\infty} \frac{1}{u} e^{-u^2} 2u du \Rightarrow 2 \int_0^{\infty} e^{-u^2} du$$

Taking square on both sides, ($\because u \rightarrow v$)

$$\begin{aligned} \left\{\gamma\left(\frac{1}{2}\right)\right\}^2 &= 4 \left(\int_0^{\infty} e^{-u^2} du \right)^2 \Rightarrow \left(2 \int_0^{\infty} e^{-u^2} du \right) \left(2 \int_0^{\infty} e^{-v^2} dv \right) \\ \left\{\gamma\left(\frac{1}{2}\right)\right\}^2 &= 4 \int_0^{\infty} \int_0^{\infty} e^{-(u^2+v^2)} du dv \end{aligned}$$

Let

$$\begin{cases} u = r \cos \theta, & v = r \sin \theta \Rightarrow u^2 + v^2 = r^2 \\ \text{As } u, v \rightarrow \infty, & \text{so } \theta = \tan^{-1}\left(\frac{v}{u}\right) \Rightarrow \theta = \frac{\pi}{2} \\ \text{And } & du dv = r dr d\theta \end{cases}$$

By putting, we get

$$\begin{aligned} \left\{\gamma\left(\frac{1}{2}\right)\right\}^2 &= 4 \int_{\theta=0}^{\frac{\pi}{2}} \int_{r=0}^{\infty} e^{-r^2} r dr d\theta \\ &= 4 \int_0^{\frac{\pi}{2}} \left\{ \int_0^{\infty} e^{-r^2} r dr \right\} d\theta \Rightarrow 4 \int_0^{\frac{\pi}{2}} \left[\frac{e^{-r^2}}{-2} \right]_0^{\infty} d\theta \\ &= \frac{4}{-2} \int_0^{\frac{\pi}{2}} \left\{ \frac{1}{\infty} - 1 \right\} d\theta \Rightarrow 2 \int_0^{\frac{\pi}{2}} d\theta \Rightarrow 2[\theta]_0^{\frac{\pi}{2}} \Rightarrow \pi \\ \text{so, } \left\{\gamma\left(\frac{1}{2}\right)\right\}^2 &= \pi \Rightarrow \gamma\left(\frac{1}{2}\right) = \sqrt{\pi} \end{aligned}$$

This is required result.

LECTURE NO. 50

Evaluating Some Improper Integrals by Gamma Function:

Question: Evaluate

$$(i) \int_0^{\infty} x^m e^{-ax^n} dx \quad (ii) \int_0^{\infty} e^{-\alpha\lambda^2} \cos\beta\lambda d\lambda$$

Solution: (i) let

$$I = \int_0^{\infty} x^m e^{-ax^n} dx$$

Let

$$\begin{cases} ax^n = u \Rightarrow x^n = \frac{u}{a} \\ du = nax^{n-1}dx \Rightarrow x = \left(\frac{u}{a}\right)^{\frac{1}{n}} \\ \text{As } x \rightarrow 0 \Rightarrow u = 0, \quad x \rightarrow \infty \Rightarrow u = \infty \end{cases}$$

By putting, we get

$$\begin{aligned} I &= \int_0^{\infty} \left\{ \left(\frac{u}{a}\right)^{\frac{1}{n}} \right\}^m e^{-u} \cdot d\left\{ \frac{u}{a} \right\}^{\frac{1}{n}} \\ I &= \frac{1}{a^{\frac{m}{n}}} \int_0^{\infty} u^{m/n} e^{-u} \left\{ \frac{1}{n} \left(\frac{u}{a}\right)^{\frac{1-n}{n}} \cdot \frac{1}{a} du \right\} \end{aligned}$$

By solving and simplifying, we have

$$I = \frac{1}{a^{\frac{m}{n} + \frac{1-n}{n} + 1}} \int_0^{\infty} \frac{1}{n} u^{\frac{m+1}{n}-1} e^{-u} du \Rightarrow \frac{1}{na^{\frac{m+1}{n}}} \gamma\left(\frac{m+1}{n}\right)$$

This is our required result.

(ii): *Do Your Self.*

LECTURE NO. 51

Inverse Linear Motion:

A particular is attracted toward a fixed point with a force inversely proportional to its instantaneous distance from fixed point. If the particle is released from rest, then find the time for it to reach fixed point.

Proof: Let at $t = 0$, the particle is at rest i.e. $v = 0$ and be at position $x = a$ and attracted towards $x = 0$. Then by given statement and by Newton's law;

$$m \frac{d^2x}{dt^2} = -\frac{k}{x} \dots (1)$$

Here m is the mass of particle and k is constant of proportionality.

$$m \frac{d^2x}{dt^2} = m \frac{d}{dt} \left(\frac{dv}{dt} \right) \Rightarrow m \frac{dv}{dt} \Rightarrow m \frac{dv}{dx} \frac{dx}{dt} \Rightarrow mv \frac{dv}{dx} = -\frac{k}{2}$$

Separating the variables;

$$mv dv = -k \frac{dx}{x}$$

Integrating;

$$m \int v dv = -k \int \frac{dx}{x} \Rightarrow m \frac{v^2}{2} = -k \ln x + c \dots (2)$$

$$\therefore v = 0 \text{ at } x = a \Rightarrow 0 = -k \ln a + c \Rightarrow c = k \ln a$$

By putting, equation (2) becomes,

$$m \frac{v^2}{2} = k \ln a - k \ln x = k \ln \left(\frac{a}{x} \right) \Rightarrow v^2 = \frac{2k}{m} \ln \left(\frac{a}{x} \right)$$

$$v = \sqrt{\frac{2k}{m}} \sqrt{\ln \left(\frac{a}{x} \right)} \Rightarrow \frac{dx}{dt} = - \sqrt{\frac{2k}{m}} \sqrt{\ln \left(\frac{a}{x} \right)} \dots (3)$$

Here – *ve* sign is taken because ‘*x*’ decreases as ‘*t*’ increases. So,

$$\sqrt{\frac{m}{2k}} \int_{x=a}^{x=0} \frac{dx}{\sqrt{\ln \left(\frac{a}{x} \right)}} = - \int_{t=0}^{t=T} dt \Rightarrow T = \sqrt{\frac{m}{2k}} \int_{x=a}^{x=0} \frac{dx}{\sqrt{\ln \left(\frac{a}{x} \right)}} \dots (4)$$

Let

$$\begin{cases} \ln \left(\frac{a}{x} \right) = u \Rightarrow e^u = \frac{a}{x} \Rightarrow x = ae^{-u} \Rightarrow dx = -ae^{-u} du \\ \text{When } x = 0 \Rightarrow u \rightarrow \infty, \quad x \rightarrow a \Rightarrow \ln \left(\frac{a}{a} \right) = \ln(1) = 0 = u \end{cases}$$

By putting, we get

$$T = \sqrt{\frac{m}{2k}} \int_0^{\infty} u^{-\frac{1}{2}} (-ae^{-u}) du \Rightarrow \sqrt{\frac{m}{2k}} \int_0^{\infty} u^{-\frac{1}{2}} e^{-u} du$$

$$T = \sqrt{\frac{m}{2k}} \cdot \gamma \left(\frac{1}{2} \right) = \sqrt{\frac{m}{2k}} \sqrt{\pi} \Rightarrow \sqrt{\frac{m\pi}{2k}}$$

This is our required result.

LECTURE NO. 52

Beta Function (Definition and Some Properties):

It is defined and given by;

$$B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx ; \quad m, n > 0$$

Let

$$\begin{cases} x = 1 - y \Rightarrow y = 1 - x \\ x = 0 \Rightarrow y = 1, \quad x = 1 \Rightarrow y = 0 \\ \therefore dx = -dy \end{cases}$$

By putting, we get

$$B(m, n) = \int_0^1 (1-y)^{m-1} y^{n-1} (-dy) \Rightarrow \int_0^1 y^{n-1} (1-y)^{m-1} dy$$

$$B(m, n) = \int_0^1 x^{n-1} (1-x)^{m-1} dx$$

Question: Prove that

$$B(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cdot \cos^{2n-1} \theta d\theta$$

Proof: Since,

$$B(m, n) = 2 \int_0^{\pi/2} x^{m-1} (1-x)^{n-1} dx$$

Let

$$\begin{cases} x = \sin^2 \theta \Rightarrow dx = 2 \sin \theta \cos \theta \\ x = 0 \Rightarrow \sin^2 \theta = 0, x = 1 \Rightarrow \sin^2 \theta = 1 \Rightarrow \theta = \frac{\pi}{2} \end{cases}$$

By putting, we get

$$\begin{aligned} B(m,n) &= \int_0^{\pi/2} (\sin^2 \theta)^{m-1} (1 - \sin^2 \theta)^{n-1} 2 \sin \theta \cos \theta \, d\theta \\ B(m,n) &= 2 \int_0^{\pi/2} (\sin^2 \theta)^{m-1} (\cos^2 \theta)^{n-1} \sin \theta \cos \theta \, d\theta \\ B(m,n) &= 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cdot \cos^{2n-1} \theta \, d\theta \end{aligned}$$

Hence proved our require result.

LECTURE NO. 53

Relation between Beta and Gamma Function:

Prove that

$$B(m,n) = \frac{\gamma(m)\gamma(n)}{\gamma(m+n)}$$

Proof: Since

$$\gamma(m) = \int_0^\infty x^{m-1} e^{-x} \, dx ; \quad n > 0.$$

Let $\{x^2 = u \Rightarrow du = 2x \, dx\}$

$$\gamma(m) = \int_0^\infty (x^2)^{m-1} e^{-x^2} 2x \, dx \Rightarrow 2 \int_0^\infty x^{2m-1} e^{-x^2} \, dx$$

Similarly,

$$\begin{aligned} \gamma(n) &= 2 \int_0^\infty y^{2n-1} e^{-y^2} \, dy \\ \gamma(m)\gamma(n) &= 4 \int_0^\infty \int_0^\infty x^{2m-1} y^{2n-1} e^{-(x^2+y^2)} \, dx \, dy \end{aligned}$$

Let

$$\begin{cases} x = r \cos \theta, y = r \sin \theta, x^2 + y^2 = r^2 \\ \theta = \tan^{-1} \left(\frac{y}{x} \right) \\ dx \, dy = r \, dr \, d\theta \end{cases}$$

By putting, we get

$$\begin{aligned} \gamma(m)\gamma(n) &= 4 \int_0^{\pi/2} \int_0^\infty (r^{2(m+n)-1} e^{-r^2} \cos^{2m-1} \theta \sin^{2n-1} \theta) \, d\theta \, dr \\ \gamma(m)\gamma(n) &= 2 \int_0^{\pi/2} (\cos^{2m-1} \theta \sin^{2n-1} \theta \, d\theta) \left(2 \int_0^\infty r^{2(m+n)-1} e^{-r^2} \, dr \right) \\ \gamma(m)\gamma(n) &= B(m,n) \gamma(m+n) \\ B(m,n) &= \frac{\gamma(m)\gamma(n)}{\gamma(m+n)} \end{aligned}$$

For better understanding, [WATCH](#) lecture.

LECTURE NO. 54

Legendre’s Duplication Formula for Gamma Function:

By evaluating the integrals,

$$I = \int_0^{\pi/2} \sin^{2p} x \, dx \quad \text{and} \quad J = \int_0^{\pi/2} \sin^{2p}(2x) \, dx$$

Derive the Legendre’s duplication formula for Gamma function.

Proof: As given

$$I = \int_0^{\pi/2} \sin^{2p} x (\cos x)^0 \, dx \quad \dots (1)$$

Comparing with,

$$\int_0^{\pi/2} \sin^{2m-1} \theta . \cos^{2n-1} x \, dx = \frac{\gamma(m)\gamma(n)}{2\gamma(m+n)} \quad \dots (2)$$

We have,

$$2p = 2m - 1 \Rightarrow m = p + \frac{1}{2} \quad \& \quad 2n - 1 = 0 \Rightarrow n = \frac{1}{2}$$

So, equation (2) can be written as,

$$\frac{\gamma\left(p+\frac{1}{2}\right)\gamma\left(\frac{1}{2}\right)}{2\gamma(p+1)} = \frac{\sqrt{\pi}\,\gamma\left(p+\frac{1}{2}\right)}{2p\,\gamma(p)} = I \quad \dots (A)$$

LECTURE NO. 55

Now for

$$J = \int_0^{\frac{\pi}{2}} \sin^{2p}(2x) \, dx$$

Let $\left\{2x = y \Rightarrow dx = \frac{1}{2} dy ; y = \pi\right\}$ by putting, we get

$$= \frac{1}{2} \int_0^{\pi} (\sin 2y)^{2p} \, dy \Rightarrow \frac{2}{2} \int_0^{\pi/2} \sin^{2p} x \, dx \Rightarrow \int_0^{\pi/2} \sin^{2p} x \, dx = I$$

Again

$$J = \int_0^{\frac{\pi}{2}} \sin^{2p}(2x) \, dx \Rightarrow \int_0^{\frac{\pi}{2}} (\sin 2x)^{2p} \, dx \Rightarrow \int_0^{\frac{\pi}{2}} (2\sin x \cos x)^{2p} \, dx$$

$$J = 2^{2p} \int_0^{\frac{\pi}{2}} \sin^{2p} x . \cos^{2p} x \, dx \Rightarrow \frac{2^{2p} \gamma\left(p+\frac{1}{2}\right)\gamma\left(p+\frac{1}{2}\right)}{2\gamma(2p+1)} \quad \dots (B)$$

$\therefore I = J$

$$\frac{\sqrt{\pi}\,\gamma\left(p+\frac{1}{2}\right)}{2p\,\gamma(p)} = \frac{2^{2p}\,\gamma\left(p+\frac{1}{2}\right)\gamma\left(p+\frac{1}{2}\right)}{2\gamma(2p+1)}$$

By solving we get,

$$\gamma(2p) = 2^{2p-1} (\pi)^{-1/2} \gamma\left(p+\frac{1}{2}\right) \gamma(p)$$

This is our required result.

LECTURE NO. 56

The Walli’s Product Formula (By Beta Function):

Prove that

$$\int_0^{\frac{\pi}{2}} \sin^p \theta d\theta = \int_0^{\frac{\pi}{2}} \cos^p \theta d\theta = \begin{cases} \frac{1.3.5 \dots (p-1)}{2.4.6 \dots p} \cdot \frac{\pi}{2}; & \text{if } p \text{ is even} \\ \frac{2.4.6 \dots (p-1)}{1.3.5 \dots p}; & \text{if } p \text{ is odd} \end{cases}$$

Solution: Since

$$B(m, n) = \frac{\gamma(m)\gamma(n)}{\gamma(m+n)} = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cdot \cos^{2n-1} \theta d\theta$$

$$\frac{\gamma(m)\gamma(n)}{2\gamma(m+n)} = \int_0^{\pi/2} \sin^{2m-1} \theta \cdot \cos^{2n-1} \theta d\theta \quad \dots (1)$$

Let $\left\{ 2m-1 = p \text{ and } 2n-1 = 0 \Rightarrow m = (p+1)\frac{1}{2}, n = \frac{1}{2} \right\}$ by putting, we get

$$\int_0^{\frac{\pi}{2}} \sin^p \theta d\theta = \frac{\gamma\left(\frac{1}{2}(p+1)\right)\gamma\left(\frac{1}{2}\right)}{2\gamma\left(\frac{1}{2}(p+2)\right)} \quad \dots (2)$$

(i): If p is even $\Rightarrow p = 2l, l \in N$, then equation (2) becomes

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \sin^p \theta d\theta &= \frac{\gamma\left(l + \frac{1}{2}\right)\gamma\left(\frac{1}{2}\right)}{2\gamma(l+1)} \\ &= \frac{\left(l - \frac{1}{2}\right)\left(l - \frac{3}{2}\right) \dots \frac{1}{2}\gamma\left(\frac{1}{2}\right)\gamma\left(\frac{1}{2}\right)}{2 \cdot l(l-1) \dots 1} \\ &= \frac{(2l-1)(2l-3) \dots 1 \left\{ \gamma\left(\frac{1}{2}\right)^2 \right\}}{2^l \cdot 2l(l-1) \dots 3 \cdot 2 \cdot 1} \\ &= \frac{(2l-1)(2l-3) \dots 5 \cdot 3 \cdot 1 (\sqrt{\pi})^2}{2l(2l-2) \dots 6 \cdot 4 \cdot 2} \\ &= \frac{1.3.5 \dots (2l-3)(2l-1) \pi}{2.4.6 \dots (2l-2)2l} \cdot \frac{\pi}{2} \end{aligned}$$

(ii): If p is odd $\Rightarrow p = 2k+1, k \in N$

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \sin^p \theta d\theta &= \frac{\gamma\left[\frac{1}{2}(2k+1+1)\right]\gamma\left(\frac{1}{2}\right)}{2\gamma\left[\frac{1}{2}(2k+3)\right]} = \frac{\gamma(k+1)\gamma\left(\frac{1}{2}\right)}{2\gamma\left(k+\frac{3}{2}\right)} \\ &= \frac{k(k-1)(k-2) \dots 3 \cdot 2 \cdot 1 \sqrt{\pi}}{2\left[\left(k+\frac{1}{2}\right)\left(k-\frac{1}{2}\right) \dots \frac{1}{2}\gamma\left(\frac{1}{2}\right)\right]} = \frac{k(k-1) \dots 3 \cdot 2 \cdot 1 (2^{k+1})}{2(2k+1)(2k-1) \dots 5 \cdot 3 \cdot 1} \\ &= 2k(2k-2) \dots 6 \cdot 4 \cdot 2 \Rightarrow 2.4.6 \dots (2k-2)2k \end{aligned}$$

This is our required result.

LECTURE NO. 57

Fourier Integral (Definition and Related Theorem):

Since Fourier series of a function $f(x)$ in $(-l, l)$ is defined and given by;

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right) \right]$$

Where

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx \quad \& \quad b_n = \frac{1}{l} \int_{-l}^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx; \quad n = 1, 2, 3, \dots$$

What if $l \rightarrow \infty$, then in this case, the Fourier series becomes the Fourier Integral.

Fourier Integral Theorem:

Let $f(x)$ be a function define on $(-\infty, \infty)$ and satisfying;

(i): $f(x)$ and $f'(x)$ are piecewise continuous in every finite interval.

(ii): $f(x)$ is absolutely integegerable in $(-\infty, \infty)$ then

$$\int_0^2 \{A(\alpha)\cos\alpha x + B(\alpha)\sin\alpha x\} d\alpha = \begin{cases} f(x) \text{ if } x \text{ is a point of continuity} \\ \frac{f(x+0) + f(x-0)}{2} \text{ if disconti.} \end{cases}$$

The L.H.S side is called Fourier Integral Expansion of $f(x)$.

LECTURE NO. 58

Some Equivalent Forms of Fourier Integral:

$$f(x) = \int_{\alpha=0}^{\infty} \{A(\alpha)\cos\alpha x + B(\alpha)\sin\alpha x\} d\alpha \quad \dots (p)$$

$$f(x) = \frac{1}{\pi} \int_{\alpha=0}^{\infty} \int_{u=-\infty}^{+\infty} f(u) \cos\alpha(x-u) du d\alpha \quad \dots (q)$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(u) e^{i\alpha(x-u)} du d\alpha \quad \dots (r)$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\alpha x} d\alpha \int_{-\infty}^{\infty} f(u) e^{-i\alpha u} du \quad \dots (s)$$

All these forms imply each other.

Question: Prove that $(p \Rightarrow q)$

$$\int_0^{\infty} \{A(\alpha)\cos\alpha x + B(\alpha)\sin\alpha x\} d\alpha \Rightarrow \frac{1}{\pi} \int_{\alpha=0}^{\infty} \int_{u=-\infty}^{+\infty} f(u) \cos\alpha(x-u) du d\alpha$$

Where

$$A(\alpha) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \cos\alpha x dx \quad \& \quad B(\alpha) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \sin\alpha x dx$$

Proof: Since Equation (q)

$$\begin{aligned} f(x) &= \frac{1}{\pi} \int_{\alpha=0}^{\infty} \int_{u=-\infty}^{+\infty} f(u) \cos\alpha(x-u) du d\alpha \\ &= \frac{1}{\pi} \int_{\alpha=0}^{\infty} \int_{u=-\infty}^{+\infty} f(u) \{\cos\alpha x \cos\alpha u + \sin\alpha x \sin\alpha u\} du d\alpha \\ &= \int_{\alpha=0}^{\infty} \left[\left\{ \frac{1}{\pi} \int_{u=-\infty}^{+\infty} f(u) \cos\alpha u du \right\} \cos\alpha x + \left\{ \frac{1}{\pi} \int_{u=-\infty}^{+\infty} f(u) \sin\alpha u du \right\} \sin\alpha x \right] d\alpha \\ &= \int_{\alpha=0}^{\infty} \{A(\alpha)\cos\alpha x + B(\alpha)\sin\alpha x\} d\alpha \end{aligned}$$

Where

$$A(\alpha) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(u) \cos\alpha u du \quad \& \quad B(\alpha) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(u) \sin\alpha u du$$

(Proved $q \Rightarrow p$)

Now Equation (p)

$$f(x) = \int_{\alpha=0}^{\infty} \{A(\alpha)\cos\alpha x + B(\alpha)\sin\alpha x\}d\alpha \quad \dots (1)$$

Where

$$A(\alpha) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(u)\cos\alpha u \, du \quad \& \quad B(\alpha) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(u)\sin\alpha u \, du$$

Putting $A(\alpha)$ and $B(\alpha)$ values in equation (1),

$$\begin{aligned} f(x) &= \int_{\alpha=0}^{\infty} \left[\left\{ \frac{1}{\pi} \int_{-\infty}^{\infty} f(u)\cos\alpha u \, du \right\} \cos\alpha x + \left\{ \frac{1}{\pi} \int_{-\infty}^{\infty} f(u)\sin\alpha u \, du \right\} \sin\alpha x \right] d\alpha \\ &= \frac{1}{\pi} \int_{\alpha=0}^{\infty} \int_{-\infty}^{\infty} f(u) [\cos\alpha x \cos\alpha u + \sin\alpha x \sin\alpha u] \, du \, d\alpha \\ f(x) &= \frac{1}{\pi} \int_{\alpha=0}^{\infty} \int_{u=-\infty}^{+\infty} f(u) \cos\alpha(x-u) \, du \, d\alpha \end{aligned}$$

(Proved $p \Rightarrow q$)

Hence Proved. (q implies p & p implies q).

LECTURE NO. 59

Fourier Transform and its Inverse (Definition):

For a function $f(x)$ defined on $(-\infty, \infty)$, Fourier Integral is defined and given by;

$$f(x) = \int_0^{\infty} \{A(\alpha)\cos\alpha x + B(\alpha)\sin\alpha x\}d\alpha$$

Where

$$A(\alpha) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x)\cos\alpha x \, dx \quad \& \quad B(\alpha) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x)\sin\alpha x \, dx$$

And one of its equivalent form is;

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\alpha x} \, d\alpha \int_{-\infty}^{\infty} f(u) e^{-i\alpha u} \, du$$

Taking

$$F(\alpha) = \int_{-\infty}^{\infty} f(u) e^{-i\alpha u} \, du \quad \dots \text{Kernal}$$

Then above relation becomes,

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\alpha) e^{i\alpha x} \, d\alpha$$

Here $F(\alpha)$ is called Fourier Transform of $f(x)$ and $f(x)$ is called Inverse Fourier Transform of $F(\alpha)$.

It can also be expressed as;

$$F(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{i\alpha x} f(x) \, d\alpha \quad \& \quad f^{-1}\{F(k)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-i\alpha x} F(x) \, d\alpha = f(x)$$

Note that in both cases/forms/formulations "product of constants should be $\frac{1}{2\pi}$."